

Using Statistics to Summarize Data Sets

I do hate averages. There is no greater mistake than to call arithmetic an exact science. There are permutations and aberrations discernible to minds entirely noble like mine; subtle variations which ordinary accountants fail to discover, hidden laws of numbers which it requires a mind like mine to perceive. For instance if you average numbers from the bottom up and then again from the top down, the result is always different.

A letter to the Mathematical Gazette (a 19th-century British mathematical journal)

The way to make sense out of raw data is to compare and contrast, to understand differences.

Gregory Bateson (in Steps to an Ecology of the Mind)

Our objective in this chapter is to develop measures that can be used to summarize a data set. These measures, formally called *statistics*, are quantities whose values are determined by the data. We study the sample mean, sample median, and sample mode. These are all statistics that measure the center or middle value of a data set. Statistics that indicate the amount of variation in the data set are also considered. We learn about what it means for a data set to be normal, and we present an empirical rule concerning such sets. We also consider data sets consisting of paired values, and we present a statistic that measures the degree to which a scatter diagram of paired values can be approximated by a straight line.

3.1 INTRODUCTION

Modern-day experiments often track certain characteristics of thousands of individuals over time. For instance, in an attempt to learn about the health consequences of certain common practices, the medical statisticians R. Doll and A. B. Hill sent questionnaires in 1951 to all doctors in the United Kingdom and received 40,000 replies. Their questionnaire dealt with age, eating habits, exercise habits, and smoking habits. These doctors were then monitored for 10 years, and the causes of death of those who died were determined. As one can

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imagine, this study resulted in huge sets of data. For instance, even if we just focus on one component of the study at a single moment of time, such as the doctors’ ages in 1951, the resulting data set of 40,000 values is vast. To obtain a feel for such a large data set, it is often necessary to summarize it by some suitably chosen measures. In this chapter, we introduce different statistics that can be used to summarize certain features of data sets.

To begin, suppose that we have in our possession sample data from some underlying population. Now, whereas in Chap. 2 we showed how to describe and portray data sets in their entirety, here we will be concerned with determining certain summary measures about the data. These summary measures are called *statistics*, where by a statistic we mean any numerical quantity whose value is determined by the data.

Definition. *Numerical quantities computed from a data set are called statistics.*

We will be concerned with statistics that describe the central tendency of the data set; that is, they describe the center of the set of data values. Three different statistics for describing this—the sample mean, sample median, and sample mode—will be presented in Secs. 3.2, 3.3, and 3.4, respectively. Once we have some idea of the center of a data set, the question naturally arises as to how much *variation* there is. That is, are most of the values close to the center, or do they vary widely about the center? In Sec. 3.5 we will discuss the sample variance and sample standard deviation, which are statistics designed to measure such variation.

In Sec. 3.6 we introduce the concept of a normal data set, which is a data set having a bell-shaped histogram. For data sets that are close to being normal, we present a rule that can be used to approximate the proportion of the data that is within a specified number of sample standard deviations from the sample mean.

In the first six sections of this chapter, we concern ourselves with data sets where each datum is a single value. However, in Sec. 3.7 we deal with paired data. That is, each data point will consist of an x value and a y value. For instance, the x value might represent the average number of cigarettes that an individual smoked per day, and the y value could be the age at which that individual died. We introduce a statistic called the *sample correlation coefficient* whose value indicates the degree to which data points having large x values also have large y values and correspondingly the degree to which those having small x values also have small y values.

The Doll–Hill study yielded the result that only about 1 in 1000 nonsmoking doctors died of lung cancer. For heavy smokers the figure was 1 in 8. In addition, death rates from heart attacks were 50 percent higher for smokers.

3.2 SAMPLE MEAN

Suppose we have a sample of n data points whose values we designate by $x_1, x_2, \dots, x_n$. One statistic for indicating the center of this data set is the *sample mean*, defined to equal the arithmetic average of the data values.

Definition. The sample mean, which we designate by $\bar{x}$ (pronounced “ x bar”), is defined by

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

■ Example 3.1

The average fuel efficiencies, as measured in miles per gallon, of cars driven in the United States in the years from 2005 to 2010 are

22.1, 22.5, 22.9, 23.7, 23.5, 23.5

Find the sample mean of this set of data.

Solution

The sample mean $\bar{x}$ is the average of the six data values. Thus,

$$\bar{x} = \frac{22.1 + 22.5 + 22.9 + 23.7 + 23.5 + 23.5}{6} = \frac{138.2}{6} = 23.003$$

Note from this example that whereas the sample mean is the average of all the data values, it need not itself be one of them. ■

Consider again the data set $x_1, x_2, \dots, x_n$. If each data value is increased by a constant amount c , then this causes the sample mean also to be increased by c . Mathematically, we can express this by saying that if

$$y_i = x_i + c \quad i = 1, \dots, n$$

then

$$\bar{y} = \bar{x} + c$$

where $\bar{y}$ and $\bar{x}$ are the sample means of the y_i and the x_i , respectively. Therefore, when it is convenient, we can compute $\bar{x}$ by first adding c to all the data values, then computing the sample mean $\bar{y}$ of the new data, and finally subtracting c from $\bar{y}$ to obtain $\bar{x}$. Since it is sometimes a lot easier to work with the transformed rather than the original data, this can greatly simplify the computation of $\bar{x}$. Our next example illustrates this point.

■ Example 3.2

The winning scores in the U.S. Masters Golf Tournament in the years from 1981 to 1990 were as follows:

280, 284, 280, 277, 282, 279, 285, 281, 283, 278

Find the sample mean of these winning scores.

Solution

Rather than directly adding the preceding numbers, first we subtract 280 from (that is, add $c = -280$ to) each one to obtain the following transformed data:

0, 4, 0, -3, 2, -1, 5, 1, 3, -2

The sample mean of these transformed data, call it $\bar{y}$, is

$$\bar{y} = \frac{0 + 4 + 0 - 3 + 2 - 1 + 5 + 1 + 3 - 2}{10} = \frac{9}{10}$$

Adding 280 to $\bar{y}$ shows that the sample mean of the original data is

$$\bar{x} = 280.9$$

If each data value is multiplied by c , then so is the sample mean. That is, if

$$y_i = cx_i \quad i = 1, \dots, n$$

then

$$\bar{y} = c\bar{x}$$

For instance, suppose that the sample mean of the height of a collection of individuals is 5.0 feet. Suppose that we now want to change the unit of measurement from feet to inches. Then since each new data value is the old value multiplied by 12, it follows that the sample mean of the new data is $12 \cdot 5 = 60$. That is, the sample mean is 60 inches.

Our next example considers the computation of the sample mean when the data are arranged in a frequency table.

■ Example 3.3

The number of suits sold daily by a women's boutique for the past 6 days has been arranged in the following frequency table:

Value	Frequency
3	2
4	1
5	3

What is the sample mean?

Solution

Since the original data set consists of the 6 values

3, 3, 4, 5, 5, 5

it follows that the sample mean is

$$\begin{aligned}\bar{x} &= \frac{3 + 3 + 4 + 5 + 5 + 5}{6} \\ &= \frac{3 \times 2 + 4 \times 1 + 5 \times 3}{6} \\ &= \frac{25}{6}\end{aligned}$$

That is, the sample mean of the number of suits sold daily is 4.25. ■

In [Example 3.3](#) we have seen that when the data are arranged in a frequency table, the sample mean can be expressed as the sum of the products of the distinct values and their frequencies, all divided by the size of the data set. This result holds in general. To see this, suppose the data are given in a frequency table that lists k distinct values $x_1, x_2, \dots, x_k$ with respective frequencies $f_1, f_2, \dots, f_k$. It follows that the data set consists of n observations, where $n = \sum_{i=1}^k f_i$ and where the value x_i appears f_i times for $i = 1, 2, \dots, k$. Hence, the sample mean for this data set is

$$\begin{aligned}\bar{x} &= \frac{x_1 + \dots + x_1 + x_2 + \dots + x_2 + \dots + x_k + \dots + x_k}{n} \\ &= \frac{f_1 x_1 + f_2 x_2 + \dots + f_k x_k}{n}\end{aligned}\tag{3.1}$$

Now, if $w_1, w_2, \dots, w_k$ are nonnegative numbers that sum to 1, then

$$w_1 x_1 + w_2 x_2 + \dots + w_k x_k$$

is said to be a *weighted average* of the values $x_1, x_2, \dots, x_k$ with w_i being the weight of x_i . For instance, suppose that $k = 2$. Now, if $w_1 = w_2 = 1/2$, then the weighted average

$$w_1 x_1 + w_2 x_2 = \frac{1}{2}x_1 + \frac{1}{2}x_2$$

is just the ordinary average of x_1 and x_2 . On the other hand, if $w_1 = 2/3$ and $w_2 = 1/3$, then the weighted average

$$w_1 x_1 + w_2 x_2 = \frac{2}{3}x_1 + \frac{1}{3}x_2$$

gives twice as much weight to x_1 as it does to x_2 .

By writing Eq. (3.1) as

$$\bar{x} = \frac{f_1}{n}x_1 + \frac{f_2}{n}x_2 + \dots + \frac{f_k}{n}x_k$$

we see that the sample mean $\bar{x}$ is a weighted average of the set of distinct values. The weight given to the value x_i is f_i/n , the proportion of the data

values that is equal to x_i . Thus, for instance, in [Example 3.3](#) we could have written that

$$\bar{x} = \frac{2}{6} \times 3 + \frac{1}{6} \times 4 + \frac{3}{6} \times 5 = \frac{25}{6}$$

■ Example 3.4

In a paper entitled “The Effects of Helmet Use on the Severity of Head Injuries in Motorcycle Accidents” (published in the *Journal of the American Statistical Association*, 1992, pp. 48–56), A. Weiss analyzed a sample of 770 similar motorcycle accidents that occurred in the Los Angeles area in 1976 and 1977. Each accident was classified according to the severity of the head injury suffered by the motorcycle operator. The classification used was as follows:

Classification of accident	Interpretation
0	No head injury
1	Minor head injury
2	Moderate head injury
3	Severe, not life-threatening
4	Severe and life-threatening
5	Critical, survival uncertain at time of accident
6	Fatal

In 331 of the accidents the operator wore a helmet, whereas in the other 439 accidents the operator did not. The following are frequency tables giving the severities of the accidents that occurred when the operator was wearing and was not wearing a helmet.

Classification	Frequency of driver with helmet	Frequency of driver without helmet
0	248	227
1	58	135
2	11	33
3	3	14
4	2	3
5	8	21
6	1	6
	331	439

Find the sample mean of the head severity classifications for those operators who wore helmets and for those who did not.

Solution

The sample mean for those wearing helmets is

$$\bar{x} = \frac{0.248 + 1.58 + 2.11 + 3.3 + 4.2 + 5.8 + 6.1}{331} = \frac{143}{331} = 0.432$$

The sample mean for those who did not wear a helmet is

$$\bar{x} = \frac{0.227 + 1.135 + 2.33 + 3.14 + 4.3 + 5.21 + 6.6}{439} = \frac{396}{439} = 0.902$$

Therefore, the data indicate that those cyclists who were wearing a helmet suffered, on average, less severe head injuries than those who were not wearing a helmet. ■

3.2.1 Deviations

Again suppose that sample data consist of the n values $x_1, \dots, x_n$ and that $\bar{x} = \sum_{i=1}^n x_i / n$ is the sample mean. The differences between each of the data values and the sample mean are called *deviations*.

Definition. The deviations are the differences between the data values and the sample mean. The value of the i th deviation is $x_i - \bar{x}$.

A useful identity is that the sum of all the deviations must equal 0. That is,

$$\sum_{i=1}^n (x_i - \bar{x}) = 0$$

That this equality is true is seen by the following argument:

$$\begin{aligned} \sum_{i=1}^n (x_i - \bar{x}) &= \sum_{i=1}^n x_i - \sum_{i=1}^n \bar{x} \\ &= n\bar{x} - n\bar{x} \\ &= 0 \end{aligned}$$

This equality states that the sum of the positive deviations from the sample mean must exactly balance the sum of the negative deviations. In physical terms, this means that if n weights of equal mass are placed on a (weightless) rod at the points $x_i, i = 1, \dots, n$, then $\bar{x}$ is the point at which the rod will be in balance. This balancing point is called the *center of gravity* (Fig. 3.1).

Historical Perspective

In the early days of sea voyages it was quite common for large portions of a ship's cargo to be either lost or damaged due to storms. To handle this potential loss, there was a standard agreement that all those having merchandise aboard the ship would contribute to pay for the value of all lost or damaged goods. The amount of money that each of them was called upon to

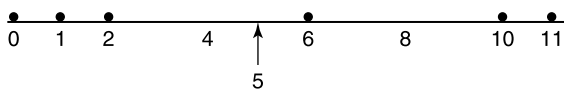


FIGURE 3.1 The center of gravity of 0, 1, 2, 6, 10, 11 is $(0 + 1 + 2 + 6 + 10 + 11)/6 = 30/6 = 5$.

pay was known as *havaria*, and from this Latin word derives our present word *average*. (Typically, if there were n shippers having damages $x_1, \dots, x_n$, then the total loss was $x_1 + \dots + x_n$ and the *havaria* for each was $(x_1 + \dots + x_n)/n$.)

■ Example 3.5

For the data of [Example 3.1](#), the deviations from the sample mean of 28.48 are

$$x_1 - \bar{x} = 28.2 - 28.48 = -0.28$$

$$x_2 - \bar{x} = 28.3 - 28.48 = -0.18$$

$$x_3 - \bar{x} = 28.4 - 28.48 = -0.08$$

$$x_4 - \bar{x} = 28.5 - 28.48 = 0.02$$

$$x_5 - \bar{x} = 29.0 - 28.48 = 0.52$$

As a check, we note that the sum of the deviations is

$$-0.28 - 0.18 - 0.08 + 0.02 + 0.52 = 0$$

PROBLEMS

- The following data represent the scores on a statistics examination of a sample of students:

87, 63, 91, 72, 80, 77, 93, 69, 75, 79, 70, 83, 94, 75, 88

What is the sample mean?

- The following data (from U.S. Department of Agriculture, *Food Consumption, Prices, and Expenditures*) give the U.S. per capita consumption (in pounds) of cheese in a sample of years.

Year	1965	1975	1985	1995	2001
Per capita consumption	10.0	14.8	23.4	26.4	30.1

Find the sample mean of the given data.

- The following data give the annual average number of inches of precipitation and the average number of days of precipitation in a sample of cities.

City	Average amount of precipitation	Average number of days
Albany, NY	35.74	134
Baltimore, MD	31.50	83
Casper, WY	11.43	95
Denver, CO	15.31	88
Fargo, ND	19.59	100
Houston, TX	44.76	105
Knoxville, TN	47.29	127
Los Angeles, CA	12.08	36
Miami, FL	57.55	129
New Orleans, LA	59.74	114
Pittsburgh, PA	36.30	154
San Antonio, TX	29.13	81
Wichita, KS	28.61	85

Source: National Oceanic and Atmospheric Administration.

- (a) Find the sample mean of the average number of inches of precipitation.
 - (b) Find the sample mean of the average number of days of precipitation.
4. Consider five numbers. Suppose the mean of the first four numbers is 14.
 - (a) If the fifth number is 24, what is the mean of all five numbers?
 - (b) If the mean of all five numbers is 24, what is the fifth number?
5. The sample mean of the weights of the adult women of town A is larger than the sample mean of the weights of the adult women of town B. Moreover, the sample mean of the weights of the adult men of town A is larger than the sample mean of the weights of the adult men of town B. Can we conclude that the sample mean of the weights of the adults of town A is larger than the sample mean of the weights of the adults of town B? Explain your answer.
6. Suppose that the sample mean of a set of 10 data points is $\bar{x} = 20$.
 - (a) If it is discovered that a data point having value 15 was incorrectly read as having value 13, what should be the revised value of $\bar{x}$?
 - (b) Suppose there is an additional data point whose value is 22. Will this increase or decrease the value of $\bar{x}$?
 - (c) Using the original data (and not the revised data in part (a)), what is the new value of $\bar{x}$ in part (b)?
7. Fill in the missing word or phrase to complete the following sentence. "If a new value is added to a set of numbers, then the sample mean will increase over what it was if the new value is —."

8. The following stem-and-leaf plot portrays the most recent 15 league bowling scores of the author of this text. Compute the sample mean.

```

18 | 2, 4, 7
17 | 0
16 | 1, 9
15 | 2, 2, 4, 8, 8
14 |
13 | 2, 1, 5, 5

```

9. Find the sample mean for this data set:

1, 2, 4, 7, 10, 12

Now find the sample means for the data sets

3, 6, 12, 21, 30, 36 and 6, 7, 9, 12, 15, 17

10. Suppose that $\bar{x}$ is the sample mean of the data set consisting of the data $x_1, \dots, x_n$. If the data are transformed according to the formula

$$y_i = ax_i + b \quad i = 1, \dots, n$$

what is the sample mean of the data set $y_1, \dots, y_n$? (In the equation, a and b are given constants.)

11. The following data give the total number of fires in Ontario, Canada, in the months of 2012:

6, 13, 5, 7, 7, 3, 7, 2, 5, 6, 9, 8

Find the sample mean of this data set.

12. The following data set specifies the total number of cars produced in the United States over a sample of years. The data are in units of 1000 cars. Find the sample mean of the number of cars sold annually in these years.

Year	1980	1985	1990	1995	2000	2002	2006
Number sold	8010	11,653	9783	11,985	12,832	12,326	11,264

Source: Statistical Abstract of the United States, 2008.

13. One-half the values of a sample are equal to 10, and the other half are equal to 20. What is the sample mean?
14. The following is a frequency table of the ages of a sample of members of a symphony for young adults.

Age value	Frequency
16	9
17	12
18	15
19	10
20	8

Find the sample mean of the given ages.

15. Half the values of a sample are equal to 10, one-sixth are equal to 20, and one-third are equal to 30. What is the sample mean?
16. There are two entrances to a parking lot. Student 1 counts the daily number of cars that pass through entrance 1, and student 2 does the same for entrance 2. Over 30 days, the data of student 1 yielded a sample mean of 122, and the data of student 2 yielded a sample mean of 160. Over these 30 days, what was the daily average number of cars that entered the parking lot?
17. A company runs two manufacturing plants. A sample of 30 engineers at plant 1 yielded a sample mean salary of \$73,600. A sample of 20 engineers at plant 2 yielded a sample mean salary of \$92,400. What is the sample mean salary for all 50 engineers?
18. Suppose that we have two distinct samples of sizes n_1 and n_2 . If the sample mean of the first sample is $\bar{x}_1$ and that of the second is $\bar{x}_2$, what is the sample mean of the combined sample of size $n_1 + n_2$?
19. The sample mean of the weights of the adult women of town A is larger than the sample mean of the weights of the adult women of town B. Moreover, the sample mean of the weights of the adult men of town A is larger than the sample mean of the weights of the adult men of town B. Can we conclude that the sample mean of the weights of the adults of town A is larger than the sample mean of the weights of the adults of town B?
20. Calculate the deviations for the data of Prob. 14 and check that they sum to 0.

3.3 SAMPLE MEDIAN

The following data represent the number of weeks after completion of a learn-to-drive course that it took a sample of seven people to obtain a driver's license:

2, 110, 5, 7, 6, 7, 3

The sample mean of this data set is $\bar{x} = 140/7 = 20$; and so six of the seven data values are quite a bit less than the sample mean, and the seventh is much greater. This points out a weakness of the sample mean as an indicator of the center of a data set—namely, its value is greatly affected by extreme data values.

A statistic that is also used to indicate the center of a data set but that is not affected by extreme values is the *sample median*, defined as the middle value when the data are ranked in order from smallest to largest. We will let m denote the sample median.

Definition. Order the data values from smallest to largest. If the number of data values is odd, then the sample median is the middle value in the ordered list; if it is even, then the sample median is the average of the two middle values.

It follows from this definition that if there are three data values, then the sample median is the second-smallest value; and if there are four, then it is the average of the second- and the third-smallest values.

■ Example 3.6

The following data represent the number of weeks it took seven individuals to obtain their driver's licenses. Find the sample median.

2, 110, 5, 7, 6, 7, 3

Solution

First arrange the data in increasing order.

2, 3, 5, 6, 7, 7, 110

Since the sample size is 7, it follows that the sample median is the fourth-smallest value. That is, the sample median number of weeks it took to obtain a driver's license is $m = 6$ weeks. ■

■ Example 3.7

The following data represent the number of days it took 6 individuals to quit smoking after completing a course designed for this purpose.

1, 2, 3, 5, 8, 100

What is the sample median?

Solution

Since the sample size is 6, the sample median is the average of the two middle values; thus,

$$m = \frac{3 + 5}{2} = 4$$

That is, the sample median is 4 days. ■

In general, for a data set of n values, the sample median is the $[(n + 1)/2]$ -smallest value when n is odd and is the average of the $(n/2)$ -smallest value and the $(n/2 + 1)$ -smallest value when n is even.

The sample mean and sample median are both useful statistics for describing the central tendency of a data set. The sample mean, being the arithmetic average, makes use of all the data values. The sample median, which makes use of only one or two middle values, is not affected by extreme values.

■ Example 3.8

The following data give the names of the National Basketball Association (NBA) individual scoring champions and their season scoring averages in each of the seasons from 2000 to 2015.

- (a) Find the sample median of the scoring averages.
 (b) Find the sample mean of the scoring averages.

1999–00	Shaquille O’Neal, L.A. Lakers	29.7
2000–01	Allen Iverson, Philadelphia 76ers	31.1
2001–02	Allen Iverson, Philadelphia 76ers	31.4
2002–03	Tracy McGrady, Orlando Magic	32.1
2003–04	Tracy McGrady, Orlando Magic	28.0
2004–05	Allen Iverson, Philadelphia 76ers	30.7
2005–06	Kobe Bryant, L.A. Lakers	35.4
2006–07	Kobe Bryant, Los Angeles Lakers	31.6
2007–08	Lebron James, Cleveland Cavaliers	30.0
2008–09	Dwyane Wade, Miami Heat	30.2
2009–10	Kevin Durant, Oklahoma Thunder	30.1
2010–11	Kevin Durant, Oklahoma Thunder	27.7
2011–12	Kevin Durant, Oklahoma Thunder	28.0
2012–13	Carmelo Anthony, New York Knicks	28.7
2013–14	Kevin Durant, Oklahoma Thunder	32.0
2014–15	Russell Westbrook, Oklahoma Thunder	28.1
2015–16	Stephen Curry, San Francisco Warriors	30.1

Solution

- (a) Since there are 17 data values, the sample median is the 9th smallest. Therefore, the sample median is

$$m = 30.1$$

- (b) The sum of all 17 values is 514.9, and so the sample mean is

$$\bar{x} = \frac{514.9}{17} \approx 30.288$$

Historical Perspective

The Dutch mathematician Christian Huyghens was one of the early developers of the theory of probability. In 1669 his brother Ludwig, after studying the mortality tables of the time, wrote to his famous older brother that “I have just been making a table showing how long people have to live. ... Live well! According to my calculations you will live to be about $56\frac{1}{2}$ and I to 55.” Christian, intrigued, also looked at the mortality tables but came up with different estimates for how long both he and his brother would live. Why? Because they were looking at different statistics. Ludwig was basing his estimates on the sample median while Christian was basing his on the sample mean!

For data sets that are roughly symmetric about their central values, the sample mean and sample median will have values close to each other. For instance, the data

4, 6, 8, 8, 9, 12, 15, 17, 19, 20, 22

are roughly symmetric about the value 12, which is the sample median. The sample mean is $\bar{x} = 140/11 = 12.73$, which is close to 12.

The question as to which of the two summarizing statistics is the more informative depends on what you are interested in learning from the data set. For instance, if a city government has a flat-rate income tax and is trying to figure out how much income it can expect, then it would be more interested in the sample mean of the income of its citizens than in the sample median (why is this?). On the other hand, if the city government were planning to construct some middle-income housing and were interested in the proportion of its citizens who would be able to afford such housing, then the sample median might be more informative (why is this?).

Although it is interesting to consider whether the sample mean or sample median is more informative in a particular situation, note that we need never restrict ourselves to a knowledge of just one of these quantities. They are both important, and thus both should always be computed when a data set is summarized.

PROBLEMS

1. The following are the total yardages of a sample of 12 municipal golf courses:

7040, 6620, 6050, 6300, 7170, 5990, 6330, 6780, 6540, 6690, 6200, 6830

(a) Find the sample median.

(b) Find the sample mean.

2. (a) Determine the sample median of the data set

14, 22, 8, 19, 15, 7, 8, 13, 20, 22, 24, 25, 11, 9, 14

(b) Increase each value in (a) by 5, and find the new sample median.

(c) Multiply each value in (a) by 3, and find the new sample median.

3. If the median of the data set $x_i, i = 1, \dots, n$, is 10, what is the median of the data set $2x_i + 3, i = 1, \dots, n$?
4. The following are the speeds of 40 cars as measured by a radar device on a city street:

22, 26, 31, 38, 27, 29, 33, 40, 36, 27, 25, 42, 28, 19, 28, 26, 33, 26, 37, 22, 31, 30, 44, 29, 25, 17, 46, 28, 31, 29, 40, 38, 26, 43, 45, 21, 29, 36, 33, 30

Find the sample median.

5. The following presents the male and female suicide rates per 100,000 population for a variety of countries.

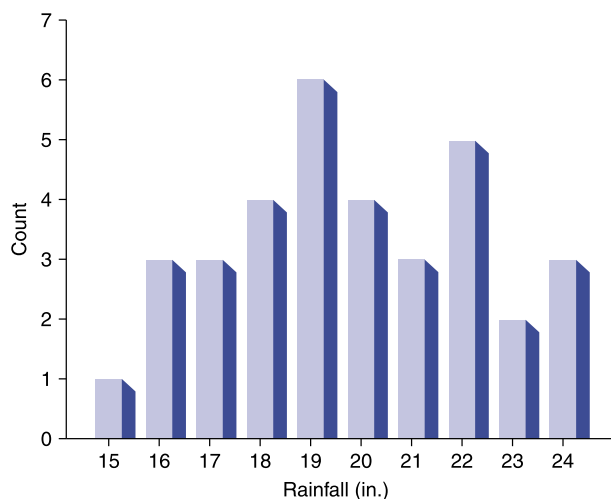
Suicide Rates per 100,000 Population

Sex	United States	Australia	Austria	Canada	Denmark	France
Female	5.4	5.1	15.8	5.4	20.6	12.7
Male	19.7	18.2	42.1	20.5	35.1	33.1

Sex	Italy	Japan	Netherlands	Poland	Sweden	U.K.	W. Germany
Female	4.3	14.9	8.1	4.4	11.5	5.7	12.0
Male	11.0	27.8	14.6	22.0	25.0	12.1	26.6

Source: World Health Organization, *World Health Statistics*.

- Find the sample median of the male suicide rates.
 - Find the sample median of the female suicide rates.
 - Find the sample mean of the male suicide rates.
 - Find the sample mean of the female suicide rates.
- Find the sample median of the average annual number of days of precipitation in the cities noted in Prob. 3 of Sec. 3.2.
 - Find the sample median of the average annual number of inches of precipitation in the cities noted in Prob. 3 of Sec. 3.2.
 - Find the sample median of the data presented in Prob. 8 of Sec. 2.3.
 - Use the table on death rates preceding Prob. 9 of Sec. 2.3 to find the sample median of the death rates due to
 - Falls
 - Poisoning
 - Drowning
 - The sample median of 10 distinct values is 5. What can you say about the new sample median if
 - An additional datum whose value is 7 is added to the data set?
 - Two additional data values—3 and 42—are added to the data set?
 - The histogram in the figure on the following page describes the annual rainfall, in inches, over the last 34 years in a certain western city. Since the raw data are not recoverable from a histogram, we cannot use them to exactly compute the value of the sample mean and sample median. Still, based on this histogram, what is the largest possible value of
 - The sample mean?
 - The sample median?
 What is the smallest possible value of
 - The sample mean?
 - The sample median?
 - The actual data follow:



15.2, 16.1, 16.5, 16.7, 17.2, 17.5, 17.7, 18.3, 18.6, 18.8, 18.9, 19.1, 19.2, 19.2, 19.6, 19.8, 19.9, 20.2, 20.3, 20.3, 20.8, 21.1, 21.4, 21.7, 22.2, 22.5, 22.5, 22.7, 22.9, 23.3, 23.6, 24.1, 24.5, 24.9

Determine the sample mean and sample median and see that they are consistent with your previous answers.

12. A total of 100 people work at company A, whereas a total of 110 work at company B. Suppose the total employee payroll is larger at company A than at company B.
 - (a) What does this imply about the sample mean of the salaries at company A with regards to the sample mean of the salaries at company B?
 - (b) What does this imply about the sample median of the salaries at company A with regards to the sample median of the salaries at company B?
13. Using the data from [Example 3.4](#), compute the sample medians of the severity of head injuries suffered by motorcycle operators who were wearing and who were not wearing helmets.
14. In the following situations, which do you think is a more informative statistic, the sample mean or the sample median?
 - (a) In order to decide whether to discontinue a bus service from Rochester to New York City, an executive studies the number of riders on a sample of days.
 - (b) To determine how present-day college-bound students compare with those of earlier years, a sample of entrance examination scores from several years is consulted.
 - (c) A lawyer representing a defendant in a jury trial is studying the IQ scores of the jurors who were selected.

- (d) You purchased your home 6 years ago in a small suburban community for \$405,000, which was both the mean and the median price for all homes sold that year in that community. However, in the last couple of years some new, more expensive homes have been built. To get an idea of the present value of your home, you study recent sales prices of homes in your community.
15. Women make up the following percentages of the workforce in the 14 occupations listed.

Occupation	Percentage women	Occupation	Percentage women
Corporate executives	36.8	Doctors	17.6
Nurses	94.3	Lawyers	18.0
Sales supervisors	30.5	Elementary school teachers	85.2
Sales workers	68.6	Postal clerks	43.5
Firefighters	1.9	Police workers	10.9
Cleaning jobs	41.5	Construction supervisors	1.6
Construction workers	2.8	Truck drivers	2.1

For these percentages find

- (a) The sample mean
(b) The sample median

It also turns out that women make up 44.4 percent of the total workforce for these occupations. Is this consistent with your answers in (a) and (b)? Explain!

16. Using data concerning the first 30 students in App. A, find the sample median and the sample mean for
- (a) Weight
(b) Cholesterol
(c) Blood pressure
17. The following table gives the median age at first marriage in the years 1992 to 2002.
- (a) Find the sample median of the men's median age.
(b) Find the sample median of the women's median age.

U.S. Median Age at First Marriage

Year	Men	Women	Year	Men	Women
2002	26.9	25.3	1996	27.1	24.8
2001	26.9	25.1	1995	26.9	24.5
2000	26.8	25.1	1994	26.7	24.5
1999	26.9	25.1	1993	26.5	24.5
1998	26.7	25.0	1992	26.5	24.4
1997	26.8	25.0			

3.3.1 Sample Percentiles

The sample median is a special type of statistic known as a *sample 100p percentile*, where p is any fraction between 0 and 1. Loosely speaking, a sample

$100p$ percentile is the value such that $100p$ percent of the data values are less than it and $100(1 - p)$ percent of the values are greater than it.

Definition. *The sample $100p$ percentile is that data value having the property that at least $100p$ percent of the data are less than or equal to it and at least $100(1 - p)$ percent of the data values are greater than or equal to it. If two data values satisfy this condition, then the sample $100p$ percentile is the arithmetic average of these values.*

Note that the sample median is the sample 50th percentile. That is, it is the sample $100p$ percentile when $p = 0.50$.

Suppose the data from a sample of size n are arranged in increasing order from smallest to largest. To determine the sample $100p$ percentile, we must determine the data value such that

1. At least np of the data values are less than or equal to it.
2. At least $n(1 - p)$ of the data values are greater than or equal to it.

Now if np is not an integer, then the only data value satisfying these requirements is the one whose position is the smallest integer greater than np . For instance, suppose we want the sample 90th percentile from a sample of size $n = 12$. Since $p = 0.9$, we have $np = 10.8$ and $n(1 - p) = 1.2$. Thus, we require those data values for which

1. At least 10.8 values are less than or equal to it (and so the data value must be in position 11 or higher).
2. At least 1.2 values are greater than or equal to it (and so it must be in position 11 or lower).

Clearly, the only data value that satisfies both requirements is the one that is in position 11, and thus this is the sample 90th percentile.

On the other hand, if np is an integer, then both the value in position np and the value in position $np + 1$ satisfy requirements (1) and (2); and so the sample $100p$ percentile value would be the average of these two data values. For instance, suppose we wanted the sample 95th percentile from a data set of $n = 20$ values. Then both the 19th and the 20th values (that is, the two largest values) will be greater than or equal to at least $np = 20(0.95) = 19$ of the values and less than or equal to at least $n(1 - p) = 1$ value. The 95th percentile is thus the average of the 19th and 20th largest values.

Summing up, we have shown the following.

To find the sample $100p$ percentile of a data set of size n

1. Arrange the data in increasing order.
2. If np is not an integer, determine the smallest integer greater than np . The data value in that position is the sample $100p$ percentile.

3. If np is an integer, then the average of the values in positions np and $np + 1$ is the sample $100p$ percentile.

■ Example 3.9

Which data value is the sample 90th percentile when the sample size is (a) 8, (b) 16, and (c) 100?

Solution

- (a) Since $0.9 \times 8 = 7.2$, which is not an integer, it follows that if the data are arranged from smallest to largest, then the sample 90th percentile value would be the 8th-smallest value (that is, the largest value).
- (b) Since $0.9 \times 16 = 14.4$, which is not an integer, it follows that the sample 90th percentile would be the 15th-smallest value.
- (c) Since $0.9 \times 100 = 90$ is an integer, the sample 90th percentile value is the average of the 90th and the 91st values when the data are arranged from smallest to largest. ■

■ Example 3.10

The universities having the largest number of months in which they ranked in the top 10 for the number of google searches over the past 114 months (as of June 2013) are as follows.

University	Number of Months in Top Ten
Harvard University	114
University of Texas, Austin	114
University of Michigan	114
Stanford University	113
University of California Los Angeles (UCLA)	111
University of California Berkeley	97
Penn State University	94
Massachusetts Institute of Technology (MIT)	66
University of Southern California (USC)	63
Ohio State University	52
Yale University	48
University of Washington	33

- (a) Find the sample 70th percentile of the number of months.
- (b) Find the sample 34th percentile of the number of months.

Solution

- (a) Because the sample size is 10 and $10 \times 0.7 = 7$, the sample 70th percentile is the average of the seventh and eighth smallest values. Equiva-

lently, it is the average of the third and fourth largest value. Hence,

$$\text{sample 70th percentile} = \frac{114 + 113}{2} = 113.5$$

- (b) Because $10 \times 0.34 = 3.4$, the sample 34th percentile is the fourth smallest value. Thus,

$$\text{sample 34th percentile} = 63$$

The sample 25th percentile, 50th percentile, and 75th percentile are known as the *quartiles*.

Definition. *The sample 25th percentile is called the first quartile. The sample 50th percentile is called the median or the second quartile. The sample 75th percentile is called the third quartile.*

The quartiles break up a data set into four parts with about 25 percent of the data values being less than the first quartile, about 25 percent being between the first and second quartiles, about 25 percent being between the second and third quartiles, and about 25 percent being larger than the third quartile.

■ Example 3.11

Find the sample quartiles for the following 18 data values, which represent the ordered values of a sample of scores from a league bowling tournament:

122, 126, 133, 140, 145, 145, 149, 150, 157, 162, 166, 175, 177, 177, 183, 188, 199, 212

Solution

Since $0.25 \times 18 = 4.5$, the sample 25th percentile is the fifth-smallest value, which is 145.

Since $0.50 \times 18 = 9$, the second quartile (or sample median) is the average of the 9th- and 10th-smallest values and so is

$$\frac{157 + 162}{2} = 159.5$$

Since $0.75 \times 18 = 13.5$, the third quartile is the 14th-smallest value, which is 177.

PROBLEMS

- Seventy-five values are arranged in increasing order. How would you determine the sample
 - 80th percentile
 - 60th percentile

- (c) 30th percentile of this data set?
2. The following table gives the number of deaths of infants per 1000 births in the 50 U.S. states in 2007. Use it to find the quartiles of the state infant death rates.

Number of Deaths of Infants per 1000 Births and Total Infant Mortality			
State Ranking			
1 Montana	4.5 (52 Total)	25 Arizona	6.7 (630 Total)
1 Vermont	4.5 (30 Total)	27 Kentucky	6.8 (378 Total)
3 Minnesota	4.7 (332 Total)	28 Florida	7.0 (1537 Total)
4 Massachusetts	4.8 (380 Total)	29 Kansas	7.2 (284 Total)
5 Iowa	5.1 (195 Total)	29 Pennsylvania	7.2 (1049 Total)
6 California	5.2 (2811 Total)	31 Illinois	7.5 (1349 Total)
6 Utah	5.2 (264 Total)	31 Missouri	7.5 (584 Total)
8 Rhode Island	5.3 (68 Total)	31 Virginia	7.5 (776 Total)
9 Connecticut	5.5 (233 Total)	34 Michigan	7.6 (984 Total)
9 Oregon	5.5 (251 Total)	34 West Virginia	7.6 (158 Total)
9 Washington	5.5 (451 Total)	36 Ohio	7.7 (1143 Total)
12 New Hampshire	5.6 (81 Total)	37 Indiana	8.0 (700 Total)
12 New Jersey	5.6 (651 Total)	37 Oklahoma	8.0 (411 Total)
12 North Dakota	5.6 (46 Total)	39 South Dakota	8.2 (93 Total)
15 Hawaii	5.7 (104 Total)	40 Arkansas	8.3 (319 Total)
15 Maine	5.7 (79 Total)	41 Maryland	8.4 (630 Total)
17 Wisconsin	6.0 (420 Total)	42 Georgia	8.5 (1181 Total)
18 New York	6.1 (1518 Total)	43 Delaware	8.6 (98 Total)
19 Idaho	6.2 (139 Total)	43 Tennessee	8.6 (687 Total)
20 Colorado	6.3 (434 Total)	45 Alabama	8.7 (516 Total)
20 New Mexico	6.3 (179 Total)	46 North Carolina	8.8 (1053 Total)
20 Texas	6.3 (2407 Total)	46 Wyoming	8.8 (60 Total)
23 Nevada	6.4 (225 Total)	48 South Carolina	9.3 (525 Total)
24 Nebraska	6.6 (173 Total)	49 Mississippi	9.8 (420 Total)
25 Alaska	6.7 (69 Total)	50 Louisiana	10.5 (684 Total)

3. Consider a data set of n values $1, 2, 3, \dots, n$. Find the value of the sample 95th percentile when
- (a) $n = 100$
- (b) $n = 101$

The following table gives the number of physicians and of dentists per 100,000 population for 12 midwestern states in 2000. Problems 4 and 5 are based on it.

State	Physician's rate	Dentist's rate
Ohio	188	56
Indiana	146	48
Illinois	206	61
Michigan	177	64
Wisconsin	177	70
Minnesota	207	70
Iowa	141	60
Missouri	186	55
North Dakota	157	55
South Dakota	129	54
Nebraska	162	71
Kansas	166	52

Source: American Medical Association, *Physician Characteristics and Distribution in the U.S.*

4. For the physician's rates per 100,000 population, find the
 - (a) Sample 40th percentile
 - (b) Sample 60th percentile
 - (c) Sample 80th percentile
5. For the dentist's rates per 100,000 population, find the
 - (a) Sample 90th percentile
 - (b) Sample 50th percentile
 - (c) Sample 10th percentile
6. Suppose the sample 100 p percentile of a set of data is 120. If we add 30 to each data value, what is the new value of the sample 100 p percentile?
7. Suppose the sample 100 p percentile of a set of data is 230. If we multiply each data value by a positive constant c , what is the new value of the sample 100 p percentile?
8. Find the sample 90th percentile of this data set:
 75, 33, 55, 21, 46, 98, 103, 88, 35, 22, 29, 73, 37, 101,
 121, 144, 133, 52, 54, 63, 21, 7
9. Find the quartiles of the following ages at death as given in obituaries of the New York Times in the two weeks preceding August 1, 2013.
 92, 90, 92, 74, 69, 80, 94, 98, 65, 96, 84, 69, 86, 91, 88
 74, 97, 85, 88, 68, 77, 94, 88, 65, 76, 75, 60
 69, 97, 92, 85, 70, 80, 93, 91, 68, 82, 78, 89
10. The following are the quartiles of a large data set:
 First quartile = 35
 Second quartile = 47
 Third quartile = 66

- (a) Give an interval in which approximately 50 percent of the data lie.
 - (b) Give a value which is greater than approximately 50 percent of the data.
 - (c) Give a value such that approximately 25 percent of the data values are greater than it.
11. A symmetric data set has its median equal to 40 and its third quartile equal to 55. What is the value of the first quartile?

3.4 SAMPLE MODE

Another indicator of central tendency is the *sample mode*, which is the data value that occurs most frequently in the data set.

■ Example 3.12

The following are the sizes of the last 8 dresses sold at a women's boutique:

8, 10, 6, 4, 10, 12, 14, 10

What is the sample mode?

Solution

The sample mode is 10, since the value of 10 occurs most frequently. ■

If no single value occurs most frequently, then all the values that occur at the highest frequency are called *modal values*. In such a situation we say that there is no unique value of the sample mode.

■ Example 3.13

The ages of 6 children at a day care center are

2, 5, 3, 5, 2, 4

What are the modal values of this data set?

Solution

Since the ages 2 and 5 both occur most frequently, both 2 and 5 are modal values. ■

■ Example 3.14

The following frequency table gives the values obtained in 30 throws of a die.

Value	Frequency
1	6
2	4
3	5
4	8
5	3
6	4

It is easy to pick out the modal value from a frequency table, since it is just that value having the largest frequency.

For these data, find the

- (a) Sample mode
- (b) Sample median
- (c) Sample mean

Solution

- (a) Since the value 4 appears with the highest frequency, the sample mode is 4.
- (b) Since there are 30 data values, the sample median is the average of the 15th- and 16th-smallest values. Since the 15th-smallest value is 3 and the 16th-smallest is 4, the sample median is 3.5.
- (c) The sample mean is

$$\bar{x} = \frac{1 \cdot 6 + 2 \cdot 4 + 3 \cdot 5 + 4 \cdot 8 + 5 \cdot 3 + 6 \cdot 4}{30} = \frac{100}{30} \approx 3.333$$

Benford's law, also called the First-Digit Law, is the observation that the first digits in many real-life sets of numerical data do not occur in equal proportions, but rather are biased towards the smaller numbers. More precisely, it states that the proportion of the data whose first digit is i , $i = 1, \dots, 9$ is approximately $\log(\frac{i+1}{i})$ with the logarithm having base 10. For instance, because $\log(2) = .301$ it says that approximately 30:1 percent of the data will have 1 as their first digit. Figure gives the proportions, according to Benford's law, of the data values having each of $1, \dots, 9$ as first digit.

Interestingly, it has been shown that this result applies to a wide variety of real life data sets, including electricity bills, street addresses, stock prices, population numbers, death rates, lengths of rivers, physical and mathematical constants, and seems to be most accurate when the data values are widely spread. The law was first published by the American astronomer Simon Newcomb in 1881. In 1938 the physicist Frank Benford tested it on data from 20 different domains and showed it was a good fit in most of these cases. Among others, he considered the surface areas of rivers, the population sizes of US cities, physical constants, and molecular weights.

PROBLEMS

- Match each statement in the left-hand column with the correct data set from the right-hand column.

- | | |
|-----------------------|------------------------|
| 1. Sample mode is 9 | A: 5, 7, 8, 10, 13, 14 |
| 2. Sample mean is 9 | B: 1, 2, 5, 9, 9, 15 |
| 3. Sample median is 9 | C: 1, 2, 9, 12, 12, 18 |

Table 3.1 Benford's Law for First Digits

First digit	Proportion of data having it as first digit
1	0.301
2	0.176
3	0.125
4	0.097
5	0.079
6	0.067
7	0.058
8	0.051
9	0.046

- Using the data from [Example 2.2](#), find the sample mode of the winning Masters Golf Tournament scores.
- Using data concerning the first 100 students in App. A, find the sample mode for
 - Weight
 - Blood pressure
 - Cholesterol
- Suppose you want to guess the salary of a bank vice president whom you have just met. If you want to have the greatest chance of being correct to the nearest \$1000, would you rather know the sample mean, the sample median, or the sample mode of the salaries of bank vice presidents?
- Construct a data set for which the sample mean is 10, the sample median is 8, and the sample mode is 6.
- If the sample mode of the data $x_i, i = 1, \dots, n$, is equal to 10, what is the sample mode of the data $y_i = 2x_i + 5, i = 1, \dots, n$?
- Joggers use a quarter-mile track around an athletic field. In a sample of 17 joggers, 1 did 2 loops, 4 did 4 loops, 5 did 6 loops, 6 did 8 loops, and 1 did 12 loops.
 - What is the sample mode of the number of loops run by these joggers?
 - What is the sample mode of the distances run by these joggers?
- The sample mean, sample median, and sample mode of the first 99 values of a data set of 198 values are all equal to 120. If the sample mean, median, and mode of the final 99 values are all equal to 100, what can you say about the sample mean of the entire data set? What can you say about the sample median? What about the sample mode?
- A multiple choice question on a physics exam, asks whether (a) 7.3316, (b) 6.2421, (c) 1.4512 or (d) 8.1818 is the density (in grams per cubed

centimeter) of a 100 percent solution of hydrogen peroxide at a temperature of 20 degrees C. Without knowing anything about hydrogen peroxide, which answer would you guess to be correct.

10. Collect data on the lengths of the drainage area of the 50 longest rivers in the world and see if they support Benford's law.

3.5 SAMPLE VARIANCE AND SAMPLE STANDARD DEVIATION

Whereas so far we have talked about statistics that measure the central tendency of a data set, we have not yet considered ones that measure its spread or variability. For instance, although the following data sets A and B have the same sample mean and sample median, there is clearly more spread in the values of B than in those of A .

A : 1, 2, 5, 6, 6 B : -40, 0, 5, 20, 35

One way of measuring the variability of a data set is to consider the deviations of the data values from a central value. The most commonly used central value for this purpose is the sample mean. If the data values are $x_1, \dots, x_n$ and the sample mean is $\bar{x} = \sum_{i=1}^n x_i / n$ then the deviation of the value x_i from the sample mean is $x_i - \bar{x}$, $i = 1, \dots, n$.

One might suppose that a natural measure of the variability of a set of data would be the average of the deviations from the mean. However, as we have shown in Sec. 3.2, $\sum_{i=1}^n (x_i - \bar{x}) = 0$. That is, the sum of the deviations from the sample mean is always equal to 0, and thus the average of the deviations from the sample mean must also be 0. However, after some additional reflection it should be clear that we really do not want to allow the positive and the negative deviations to cancel. Instead, we should be concerned about the individual deviations without regard to their signs. This can be accomplished either by considering the absolute values of the deviations or, as turns out to be more useful, by considering their squares.

The sample variance is a measure of the "average" of the squared deviations from the sample mean. However, for technical reasons (which will become clear in Chap. 8) this "average" divides the sum of the n squared deviations by the quantity $n - 1$, rather than by the usual value n .

Definition. The sample variance, call it s^2 , of the data set $x_1, \dots, x_n$ having sample mean $\bar{x} = \sum_{i=1}^n x_i / n$ is defined by

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

■ Example 3.15

Find the sample variance of data set A .

Solution

It is determined as follows:

x_i	1	2	5	6	6
$\bar{x}$	4	4	4	4	4
$x_i - \bar{x}$	-3	-2	1	2	2
$(x_i - \bar{x})^2$	9	4	1	4	4

Hence, for data set A ,

$$s^2 = \frac{9 + 4 + 1 + 4 + 4}{4} = 5.5$$

■ **Example 3.16**

Find the sample variance for data set B .

Solution

The sample mean for data set B is also $\bar{x} = 4$. Therefore, for this set, we have

x_i	-40	0	5	20	35
$x_i - \bar{x}$	-44	-4	1	16	31
$(x_i - \bar{x})^2$	1936	16	1	256	961

Thus,

$$s^2 = \frac{3170}{4} = 792.5$$

The following algebraic identity is useful for computing the sample variance by hand:

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2 \quad (3.2)$$

■ **Example 3.17**

Check that identity (3.2) holds for data set A .

Solution

Since $n = 5$ and $\bar{x} = 4$,

$$\sum_{i=1}^5 x_i^2 - n\bar{x}^2 = 1 + 4 + 25 + 36 + 36 - 5(16) = 102 - 80 = 22$$

From [Example 3.15](#),

$$\sum_{i=1}^5 (x_i - \bar{x})^2 = 9 + 4 + 1 + 4 + 4 = 22$$

and so the identity checks out. ■

Suppose that we add a constant c to each of the data values $x_1, \dots, x_n$ to obtain the new data set $y_1, \dots, y_n$, where

$$y_i = x_i + c$$

To see how this affects the value of the sample variance, recall from [Sec. 3.2](#) that

$$\bar{y} = \bar{x} + c$$

and so

$$y_i - \bar{y} = x_i + c - (\bar{x} + c) = x_i - \bar{x}$$

That is, the y deviations are equal to the x deviations, and therefore their sums of squares are equal. Thus, we have shown the following useful result.

The sample variance remains unchanged when a constant is added to each data value.

The preceding result can often be used in conjunction with the algebraic identity [\(3.2\)](#) to greatly reduce the time it takes to compute the sample variance.

■ Example 3.18

The following data give the yearly numbers of law enforcement officers killed in the United States over 10 years:

164, 165, 157, 164, 152, 147, 148, 131, 147, 155

Find the sample variance of the number killed in these years.

Solution

Rather than working directly with the given data, let us subtract the value 150 from each data item. (That is, we are adding $c = -150$ to each data value.) This results in the new data set

14, 15, 7, 14, 2, -3, -2, -19, -3, 5

Its sample mean is

$$\bar{y} = \frac{14 + 15 + 7 + 14 + 2 - 3 - 2 - 19 - 3 + 5}{10} = 3.0$$

The sum of the squares of the new data is

$$\sum_{i=1}^{10} y_i^2 = 14^2 + 15^2 + 7^2 + 14^2 + 2^2 + 3^2 + 2^2 + 19^2 + 3^2 + 5^2 = 1078$$

Therefore, using the algebraic identity (3.2) shows that

$$\sum_{i=1}^{10} (y_i - \bar{y})^2 = 1078 - 10(9) = 988$$

Hence, the sample variance of the revised data, which is equal to the sample variance of the original data, is

$$s^2 = \frac{988}{9} \approx 109.78$$

The positive square root of the sample variance is called the *sample standard deviation*.

Definition. The quantity s , defined by

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$$

is called the sample standard deviation.

The sample standard deviation is measured in the same units as the original data. That is, for instance, if the data are in feet, then the sample variance will be expressed in units of square feet and the sample standard deviation in units of feet.

If each data value $x_i, i = 1, \dots, n$, is multiplied by a constant c to obtain the new data set

$$y_i = cx_i \quad i = 1, \dots, n$$

then the sample variance of the y data is the sample variance of the x data multiplied by c^2 . That is,

$$s_y^2 = c^2 s_x^2$$

where s_y^2 and s_x^2 are the sample variances of the new and old data sets, respectively. Taking the square root of both sides of the preceding equation shows that the standard deviation of the y data is equal to the absolute value of c times the standard deviation of the x data, or

$$s_y = |c|s_x$$

Another indicator of the variability of a data set is the interquartile range, which is equal to the third minus the first quartile. That is, roughly speaking,

Table 3.2 Selected Percentiles on the Miller Analogies Test for Five Categories of Students

Percentile	Physical sciences	Medical school	Social sciences	Languages and literature	Law school
99	93	92	90	87	84
90	88	78	82	80	73
75	80	71	74	73	60
50	68	57	61	59	49
25	55	45	49	43	37

the interquartile range is the length of the interval in which the middle half of the data values lie.

■ Example 3.19

The Miller Analogies Test is a standardized test that is taken by a variety of students applying to graduate and professional schools. Table 3.2 presents some of the percentile scores on this examination for students, classified according to the graduate fields they are entering. For instance, Table 3.2 states that the median grade of students in the physical sciences is 68, whereas it is 49 for those applying to law school.

Determine the interquartile ranges of the scores of students in the five specified categories.

Solution

Since the interquartile range is the difference between the 75th and the 25th sample percentiles, it follows that its value is

$$80 - 55 = 25 \quad \text{for scores of physical science students}$$

$$71 - 45 = 26 \quad \text{for scores of medical school students}$$

$$74 - 49 = 25 \quad \text{for scores of social science students}$$

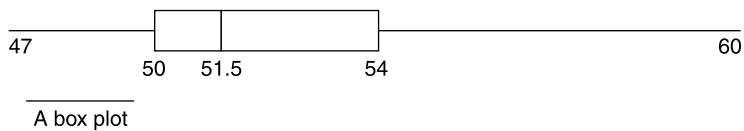
$$73 - 43 = 30 \quad \text{for scores of language and literature students}$$

$$60 - 37 = 23 \quad \text{for scores of law school students}$$

A *box plot* is often used to plot some of the summarizing statistics of a data set. A straight-line segment stretching from the smallest to the largest data value is drawn on a horizontal axis; imposed on the line is a “box,” which starts at the first and continues to the third quartile, with the value of the second quartile indicated by a vertical line. For instance, the following frequency table gives the starting salaries of a sample of 42 graduating seniors of a liberal arts college.

Starting salary	Frequency
47	4
48	1
49	3
50	5
51	8
52	10
53	0
54	5
56	2
57	3
60	1

The salaries go from a low of 47 to a high of 60. The value of the first quartile (equal to the value of the 11th smallest on the list) is 50; the value of the second quartile (equal to the average of the 21st- and 22nd-smallest values) is 51.5; and the value of the third quartile (equal to the value of the 32nd smallest on the list) is 54. The box plot for this data set is as follows.



PROBLEMS

1. A random group of 12 high school juniors were asked to estimate the average number of hours they study each week. The following give these hours along with the student's grade point average.

Hours	GPA
6	2.8
14	3.2
3	3.1
22	3.6
9	3.0
11	3.3
12	3.4
5	2.7
18	3.1
24	3.8
15	3.0
17	3.9

- (a) Find the sample mean of hours studied.

- (b) Find the sample variance of hours studied.
 - (c) Find the sample mean of GPA.
 - (d) Find the sample variance of GPA.
2. You are given these data sets:
- A: 66, 68, 71, 72, 72, 75 B: 2, 5, 9, 10, 10, 16
- (a) Which one appears to have the larger sample variance?
 - (b) Determine the sample variance of data set A.
 - (c) Determine the sample variance of data set B.
3. The Masters Golf Tournament and the U.S. Open are the two most prestigious golf tournaments in the United States. The Masters is always played on the Augusta National golf course, whereas the U.S. Open is played on different courses in different years. As a result, one might expect the sample variance of the winning scores in the U.S. Open to be higher than that of the winning scores in the Masters. To check whether this is so, we have collected the winning scores in both tournaments for 1981 to 1990.

Winning score										
Tournament	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990
U.S. Open	273	282	280	276	279	279	277	278	278	280
Masters	280	284	280	277	282	279	285	281	283	278

- (a) Compute the sample variance of the winning scores in the U.S. Open tournament.
- (b) Compute the sample variance of the winning scores in the Masters tournament.

The following table gives the numbers of physicians and dentists in Japan in the even-numbered years between 1984 and 2000. Problems 4 and 5 are based on this table.

Number of Physicians and Dentists (1984–2000)		
	Physicians	Dentists
1984	173,452	61,283
1986	183,129	64,904
1988	193,682	68,692
1990	203,797	72,087
1992	211,498	75,628
1994	220,853	79,091
1996	230,297	83,403
1998	236,933	85,669
2000	243,201	88,410

4. Just by eyeballing, estimate the ratio of the sample variance of the yearly number of physicians to the sample variance of the yearly number of dentists.

5. Find the actual value of the ratio in problem 4.
6. An individual needing automobile insurance requested quotes from 10 different insurers for identical coverage and received the following values (amounts are six month premiums in dollars):

720, 880, 630, 590, 1140, 908, 677, 720, 1260, 800

Find

- (a) The sample mean
- (b) The sample median
- (c) The sample standard deviation
7. If the sample variance of a set of 10 numbers is 44 and the sample variance of a different set of 10 numbers is 88, what can be said about the sample variance of the combined set of all 20 numbers.
8. The following are the heights and starting salaries of 12 law school classmates whose law school examination scores were roughly the same.

Height	Salary
64	91
65	94
66	88
67	103
69	77
70	96
72	105
72	88
74	122
74	102
75	90
76	114

Find the sample variance of the heights of the 12 students.

9. In the previous problem, find the sample variance of the starting salaries of the 12 students.
10. If s^2 is the sample variance of the data $x_i, i = 1, \dots, n$, what is the sample variance of the data $ax_i + b, i = 1, \dots, n$, when a and b are given constants?
11. Compute the sample variance and sample standard deviation of the following data sets:
 - (a) 1, 2, 3, 4, 5
 - (b) 6, 7, 8, 9, 10
 - (c) 11, 12, 13, 14, 15
 - (d) 2, 4, 6, 8, 10
 - (e) 10, 20, 30, 40, 50
12. On the U.S. side of the U.S.–Canada border, temperatures are measured in degrees Fahrenheit, whereas on the Canadian side they are measured in degrees Celsius (also called Centigrade). Suppose that during the month of January the sample mean of the temperatures, as

recorded on the U.S. side of the border, was 40°F with a sample variance of 12.

Use the formula for converting a Fahrenheit temperature to a Celsius temperature

$$C = \frac{5}{9}(F - 32)$$

to find

- (a) The sample mean recorded by the Canadians
- (b) The sample variance recorded by the Canadians

13. Compute the sample mean and sample variance of the systolic blood pressures of the first 50 students of the data set of App. A. Now do the same with the last 50 students of this data set. Compare your answers. Comment on the results of this comparison. Do you find it surprising?
14. If s is the sample standard deviation of the data $x_i, i = 1, \dots, n$, what is the sample standard deviation of $ax_i + b, i = 1, \dots, n$? In this problem, a and b are given constants.
15. The following are the number of electric bikes sold, in units of 1,000, in the United States in the years from 2012 through 2016

70, 159, 276, 350, 400

Find the sample standard deviation of the yearly number sold.

16. Find the sample standard deviation of the data set given by the following frequency table:

Value	Frequency	Value	Frequency
3	1	5	3
4	2	6	2

17. The following data represent the acidity of 40 successive rainfalls in the state of Minnesota. The acidity is measured on a pH scale, which varies from 1 (very acidic) to 7 (neutral).
 3.71, 4.23, 4.16, 2.98, 3.23, 4.67, 3.99, 5.04, 4.55, 3.24, 2.80, 3.44, 3.27, 2.66, 2.95, 4.70, 5.12, 3.77, 3.12, 2.38, 4.57, 3.88, 2.97, 3.70, 2.53, 2.67, 4.12, 4.80, 3.55, 3.86, 2.51, 3.33, 3.85, 2.35, 3.12, 4.39, 5.09, 3.38, 2.73, 3.07
 - (a) Find the sample standard deviation.
 - (b) Find the range.
 - (c) Find the interquartile range.
18. Consider the following two data sets:
 A: 4.5, 0, 5.1, 5.0, 10, 5.2 B: 0.4, 0.1, 9, 0, 10, 9.5
 - (a) Determine the range for each data set.
 - (b) Determine the sample standard deviation for each data set.
 - (c) Determine the interquartile range for each data set.

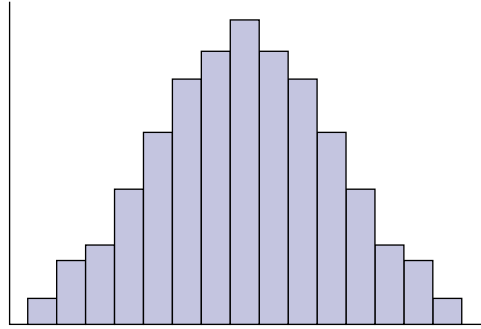


FIGURE 3.2 Histogram of a normal data set.

3.6 NORMAL DATA SETS AND THE EMPIRICAL RULE

Many of the large data sets one encounters in practice have histograms that are similar in shape. These histograms are often symmetric about their point of highest frequency and then decrease on both sides of this point in a bell-shaped fashion. Such data sets are said to be *normal*, and their histograms are called *normal histograms*.

Definition. A data set is said to be normal if a histogram describing it has the following properties:

1. It is highest at the middle interval.
2. Moving from the middle interval in either direction, the height decreases in such a way that the entire histogram is bell-shaped.
3. The histogram is symmetric about its middle interval.

Figure 3.2 shows the histogram of a normal data set.

If the histogram of a data set is close to being a normal histogram, then we say that the data set is *approximately normal*. For instance, the histogram given in Fig. 3.3 is from an approximately normal data set, whereas the ones presented in Figs. 3.4 and 3.5 are not (since each is too nonsymmetric). Any data set that is not approximately symmetric about its sample median is said to be *skewed*. It is called *skewed to the right* if it has a long tail to the right and *skewed to the left* if it has a long tail to the left. Thus the data set presented in Fig. 3.4 is skewed to the left, and the one of Fig. 3.5 is skewed to the right.

It follows from the symmetry of the normal histogram that a data set that is approximately normal will have its sample mean and sample median approximately equal.

Suppose that $\bar{x}$ and s are the sample mean and sample standard deviation, respectively, of an approximately normal data set. The following rule, known as

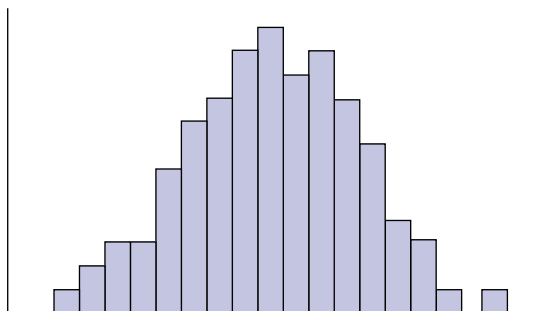


FIGURE 3.3 Histogram of an approximately normal data set.

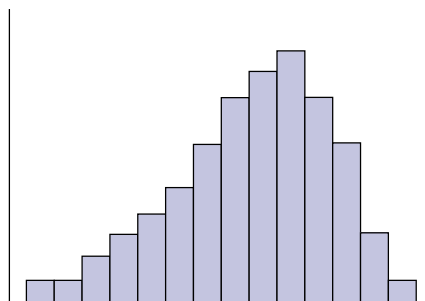


FIGURE 3.4 Histogram of a data set skewed to the left.

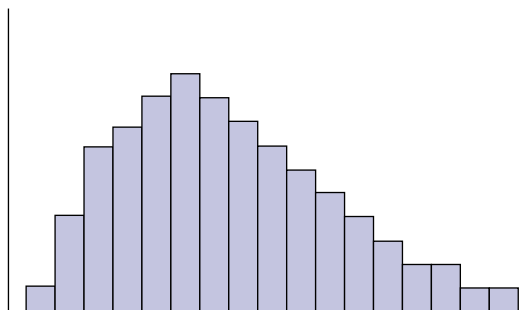


FIGURE 3.5 Histogram of a data set skewed to the right.

the *empirical rule*, specifies the approximate proportions of the data observations that are within s , $2s$, and $3s$ of the sample mean $\bar{x}$.

Empirical Rule

If a data set is approximately normal with sample mean $\bar{x}$ and sample standard deviation s , then the following are true.

1. Approximately 68 percent of the observations lie within
 $\bar{x} \pm s$
2. Approximately 95 percent of the observations lie within
 $\bar{x} \pm 2s$
3. Approximately 99.7 percent of the observations lie within
 $\bar{x} \pm 3s$

■ Example 3.20

The scores of 25 students on a history examination are listed on the following stem-and-leaf plot.

9		0, 0, 4
8		3, 4, 4, 6, 6, 9
7		0, 0, 3, 5, 5, 8, 9
6		2, 2, 4, 5, 7
5		0, 3, 5, 8

By standing this figure on its side (or, equivalently, by turning the textbook), we can see that the corresponding histogram is approximately normal. Use it to assess the empirical rule.

Solution

A calculation yields that the sample mean and sample standard deviation of the data are

$$\bar{x} = 73.68 \quad \text{and} \quad s = 12.80$$

The empirical rule states that approximately 68 percent of the data values are between $\bar{x} - s = 60.88$ and $\bar{x} + s = 86.48$. Since 17 of the observations actually fall within 60.88 and 86.48, the actual percentage is $100(17/25) = 68$ percent. Similarly, the empirical rule states that approximately 95 percent of the data are between $\bar{x} - 2s = 48.08$ and $\bar{x} + 2s = 96.28$, whereas, in actuality, 100 percent of the data fall in this range. ■

A data set that is obtained by sampling from a population that is itself made up of subpopulations of different types is usually not normal. Rather, the histogram from such a data set often appears to resemble a combining, or superposition, of normal histograms and thus will often have more than one local peak or hump. Because the histogram will be higher at these local peaks than at their neighboring values, these peaks are similar to modes. A data set whose histogram has two local peaks is said to be *bimodal*. The data set represented in Fig. 3.6 is bimodal.

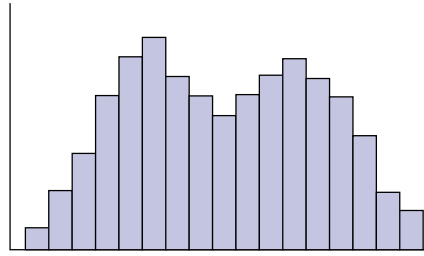


FIGURE 3.6 Histogram of a bimodal data set.

Since a stem-and-leaf plot can be regarded as a histogram lying on its side, it is useful in showing us whether a data set is approximately normal.

■ Example 3.21

The following is the stem-and-leaf plot of the weights of 200 members of a health club.

24	9
23	
22	1
21	7
20	2, 2, 5, 5, 6, 9, 9, 9
19	0, 0, 0, 0, 0, 1, 1, 2, 4, 4, 5, 8
18	0, 1, 1, 2, 2, 2, 3, 4, 4, 4, 5, 5, 5, 6, 6, 6, 6, 7, 9, 9, 9
17	1, 1, 1, 2, 3, 3, 4, 4, 4, 5, 5, 6, 6, 6, 6, 7, 7, 7, 9
16	0, 0, 1, 1, 1, 1, 2, 4, 5, 5, 6, 6, 8, 8, 8, 8
15	0, 1, 1, 1, 1, 1, 1, 5, 5, 5, 5, 6, 6, 6, 7, 7, 8, 9
14	0, 0, 0, 1, 2, 3, 4, 5, 6, 7, 7, 7, 8, 9, 9
13	0, 0, 0, 1, 1, 1, 2, 2, 2, 2, 2, 3, 3, 4, 5, 5, 6, 6, 6, 6, 7, 7, 8, 8, 8, 9, 9, 9
12	1, 1, 1, 2, 2, 2, 3, 4, 4, 5, 5, 6, 6, 6, 6, 6, 6, 7, 7, 7, 7, 8, 8, 9, 9, 9
11	0, 1, 1, 2, 2, 2, 2, 3, 3, 4, 4, 5, 5, 6, 9, 9
10	0, 2, 3, 3, 3, 4, 4, 5, 7, 7, 8
9	0, 0, 9
8	6

By standing it on its side, we see that its histogram does not appear to be approximately normal. However, it is important to note that these data consist of the weights of all members of the health club, both female and male. Since these are clearly separate populations with regard to weight, it makes sense to consider the data for each gender separately. We will now do so.

It turns out that these 200 data values are the weights of 97 women and 103 men. Separating the data for women and men results in the stem-and-leaf plots in [Figs. 3.7 and 3.8](#).

16	0, 5
15	0, 1, 1, 1, 5
14	0, 0, 1, 2, 3, 4, 6, 7, 9
13	0, 0, 1, 1, 2, 2, 2, 2, 3, 4, 5, 5, 6, 6, 6, 6, 7, 8, 8, 8, 9, 9, 9
12	1, 1, 1, 2, 2, 2, 3, 4, 4, 5, 5, 6, 6, 6, 6, 6, 6, 6, 7, 7, 7, 7, 8, 8, 9, 9
11	0, 0, 1, 1, 2, 2, 2, 2, 3, 3, 4, 4, 5, 5, 6, 9, 9
10	2, 3, 3, 3, 4, 4, 5, 7, 7, 8
9	0, 0, 9
8	6

FIGURE 3.7 Weights of 97 female health club members.

24	9
23	
22	1
21	7
20	2, 2, 5, 5, 6, 9, 9, 9
19	0, 0, 0, 0, 0, 1, 1, 2, 4, 4, 5, 8
18	0, 1, 1, 2, 2, 2, 3, 4, 4, 4, 5, 5, 5, 6, 6, 6, 6, 7, 9, 9, 9
17	1, 1, 1, 2, 3, 3, 4, 4, 4, 5, 5, 6, 6, 6, 6, 7, 7, 7, 7, 9
16	0, 1, 1, 1, 1, 2, 4, 5, 6, 6, 8, 8, 8, 8
15	1, 1, 1, 5, 5, 5, 6, 6, 6, 7, 7, 8, 9
14	0, 5, 7, 7, 8, 9
13	0, 1, 2, 3, 7
12	9

FIGURE 3.8 Weights of 103 male health club members.

As we can see from the figures, the separated data for each sex appear to be approximately normal. Let us calculate $\bar{x}_w$, s_w , $\bar{x}_m$, and s_m , the sample mean and sample standard deviation for, respectively, the women and the men.

This calculation yields

$$\bar{x}_w = 125.70 \quad \bar{x}_m = 174.69$$

$$s_w = 15.58 \quad s_m = 21.23$$

A further corroboration of the approximate normality of the two sets of separated data is provided by noting the similar values in each set of the sample mean and sample median. The sample median of the women's weights is the 49th-smallest data value, which equals 126, whereas for the men's data the sample median is the 52nd-smallest data value, which equals 174. These are quite close to the two sample means, whose values are 125.7 and 174.69.

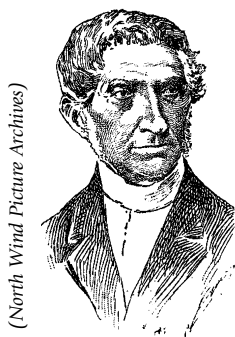
Given the values of the sample mean and sample standard deviation, it follows from the empirical rule that approximately 68 percent of the women

will weigh between 110.1 and 141.3 and approximately 95 percent of the men will weigh between 132.2 and 217.2. The actual percentages from Figs. 3.7 and 3.8 are

$$100 \times \frac{68}{97} = 70.1 \quad \text{and} \quad 100 \times \frac{101}{103} = 98.1$$

PROBLEMS

1. The daily numbers of animals treated at a certain veterinarian clinic over a 24-day period are as follows:
22, 17, 19, 31, 28, 29, 21, 33, 36, 24, 15, 28, 25, 28, 22,
27, 33, 19, 25, 28, 26, 20, 30, 32
 - (a) Plot these data in a histogram.
 - (b) Find the sample mean.
 - (c) Find the sample median.
 - (d) Is this data set approximately normal?



(North Wind Picture Archives)

Adolphe Quetelet

Historical Perspective

Quetelet and How the Normal Curve Uncovered Fraud

The Belgian social scientist and statistician Adolphe Quetelet was a great believer in the hypothesis that most data sets relating to human measurements are normal. In one study he measured the chests of 5738 Scottish soldiers, plotted the resulting data in a histogram, and concluded that it was normal.

In a later study Quetelet used the shape of the normal histogram to uncover evidence of fraud in regard to draft conscripts to the French army. He studied data concerning the heights of a huge sample of 100,000 conscripts. Plotting the data in a histogram—with class intervals of 1 inch—he found that, with the exception of three class intervals around 62 inches, the data appeared to be normal. In particular, there were fewer values in the interval from 62 to 63 inches and slightly more in the intervals from 60 to 61 and from 61 to 62 inches than would have occurred with a perfect normal fit of the data. Trying to figure out why the normal curve did not fit as well as he had supposed it would, Quetelet discovered that 62 inches was the minimum height required for soldiers in the French army. Based on this and his confidence in the widespread applicability of normal data, Quetelet concluded that some conscripts whose heights were slightly above 62 inches were “bending their knees” to appear shorter so as to avoid the draft.

For 50 years following Quetelet, that is, roughly from 1840 to 1890, it was widely believed that most data sets from homogeneous populations (that is, data that were not obviously a mixture of different populations) would appear to be normal if the sample size were sufficiently large. Whereas present-day statisticians have become somewhat skeptical about this claim, it is quite common for a data set to appear to come from a normal population. This phenomenon, which often appears in data sets originating in either the biological or the physical sciences, is partially explained by a mathematical result known as the *central limit theorem*. Indeed, the central limit theorem (studied in Chap. 7) will in itself explain why many data sets originating in the physical sciences are approximately normal. To explain why biometric data (that is, data generated by studies in biology) often appear to be normal, we will use what was originally an empirical observation noted by Francis Galton but that nowadays has a sound scientific explanation, called *regression to the mean*. Regression to the mean, in conjunction with the central limit theorem

and the passing of many generations, will yield our explanation as to why a biometric data set is often normal. The explanation will be presented in Chap. 12.

2. The following data give the injury rates per 100,000 worker-hours for a sample of 20 semiconductor firms:

1.4, 2.4, 3.7, 3.1, 2.0, 1.9, 2.5, 2.8, 2.2, 1.7, 3.1, 4.0, 2.2, 1.8,
2.6, 3.6, 2.9, 3.3, 2.0, 2.4

- (a) Plot the data in a histogram.
- (b) Is the data set roughly symmetric?
- (c) If the answer to (b) is no, is it skewed to the left or to the right?
- (d) If the answer to (b) is yes, is it approximately normal?

The following table gives the 2006 per capita consumption of milk in various countries. Problems 3 and 4 refer to this table.

Per Capita Consumption of Milk and Milk Products in
Various Countries, 2006 data

Country	Liquid milk drinks (liters)
Finland	183.9
Sweden	145.5
Ireland	129.8
Netherlands	122.9
Norway	116.7
Spain (2005)	119.1
Switzerland	112.5
United Kingdom (2005)	111.2
Australia (2005)	106.3
Canada (2005)	94.7
European Union (25 countries)	92.6
Germany	92.3
France	92.2
New Zealand (2005)	90.0
United States	83.9
Austria	80.2
Greece	69.0
Argentina (2005)	65.8
Italy	57.3
Mexico	40.7
China (2005)	8.8

Source: International Dairy Federation, Bulletin 423/2007.

- 3. Find the sample mean and sample median of the milk consumption data set.
- 4. Plot the milk consumption data in a stem and leaf plot. Is the data set approximately normal?

5. The following represent the times (in minutes) it took 22 newly hired workers to complete a standardized task:

166, 82, 175, 181, 169, 177, 180, 185, 159, 164, 170, 149, 188,
173, 170, 164, 158, 177, 173, 175, 190, 172

- Find the sample mean.
 - Find the sample median.
 - Plot the data in a histogram.
 - Is this data set approximately normal?
6. The following data give the age at inauguration of all 43 presidents of the United States.

President	Age at inauguration	President	Age at inauguration
1. Washington	57	23. B. Harrison	55
2. J. Adams	61	24. Cleveland	55
3. Jefferson	57	25. McKinley	54
4. Madison	57	26. T. Roosevelt	42
5. Monroe	58	27. Taft	51
6. J.Q. Adams	57	28. Wilson	56
7. Jackson	61	29. Harding	55
8. Van Buren	54	30. Coolidge	51
9. W. Harrison	68	31. Hoover	54
10. Tyler	51	32. F. Roosevelt	51
11. Polk	49	33. Truman	60
12. Taylor	64	34. Eisenhower	62
13. Fillmore	50	35. Kennedy	43
14. Pierce	48	36. L. Johnson	55
15. Buchanan	65	37. Nixon	56
16. Lincoln	52	38. Ford	61
17. A. Johnson	56	39. Carter	52
18. Grant	46	40. Reagan	69
19. Hayes	54	41. G. H. W. Bush	64
20. Garfield	49	42. Clinton	46
21. Arthur	50	43. G. W. Bush	54
22. Cleveland	47	44. Obama	47

- Find the sample mean and sample standard deviation of this data set.
 - Draw a histogram for the given data.
 - Do the data appear to be approximately normal?
 - If the answer to (c) is yes, give an interval that you would expect to contain approximately 95 percent of the data observations.
 - What percentage of the data lies in the interval given in part (d)?
7. For the data on the weights of female health club members presented in Fig. 3.7, the sample mean and sample standard deviation were computed to be 125.70 and 15.58, respectively. Based on the shape of Fig. 3.7

and these values, approximate the proportion of the women whose weight is between 94.54 and 156.86 pounds. What is the actual proportion?

8. A sample of 36 male coronary patients yielded the following data concerning the ages at which they suffered their first heart attacks.

7	1, 2, 4, 5
6	0, 1, 2, 2, 3, 4, 5, 7
5	0, 1, 2, 3, 3, 4, 4, 4, 5, 6, 7, 8, 9
4	1, 2, 2, 3, 4, 5, 7, 8, 9
3	7, 9

- Determine $\bar{x}$ and s .
 - From the shape of the stem-and-leaf plot, what percentage of data values would you expect to be between $\bar{x} - s$ and $\bar{x} + s$? Between $\bar{x} - 2s$ and $\bar{x} + 2s$?
 - Find the actual percentages for the intervals given in (b).
9. If the histogram is skewed to the right, which statistic will be larger—the sample mean or the sample median? (*Hint* : If you are not certain, construct a data set that is skewed to the right and then calculate the sample mean and sample median.)
10. The following data are the ages of a sample of 36 victims of violent crime in a large eastern city:

25, 16, 14, 22, 17, 20, 15, 18, 33, 52, 70, 38, 18, 13, 22, 27, 19, 23,
33, 15, 13, 62, 21, 57, 66, 16, 24, 22, 31, 17, 20, 14, 26, 30, 18, 25

- Determine the sample mean.
- Find the sample median.
- Determine the sample standard deviation.
- Does this data set appear to be approximately normal?
- What proportion of the data lies within 1 sample standard deviation of the sample mean?
- Compare your answer in (e) to the approximation provided by the empirical rule.

3.7 SAMPLE CORRELATION COEFFICIENT

Consider the data set of paired values $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. In this section we will present a statistic, called the *sample correlation coefficient*, that measures the degree to which larger x values go with larger y values and smaller x values go with smaller y values.

The data in Table 3.3 represent the average daily number of cigarettes smoked (the x variable) and the number of free radicals (the y variable), in a suitable unit, found in the lungs of 10 smokers. (A free radical is a single atom of oxygen. It is believed to be potentially harmful because it is highly reactive and has

Table 3.3 Cigarette Smoking and Free Radicals		
Person	Number of cigarettes smoked	Free radicals
1	18	202
2	32	644
3	25	411
4	60	755
5	12	144
6	25	302
7	50	512
8	15	223
9	22	183
10	30	375

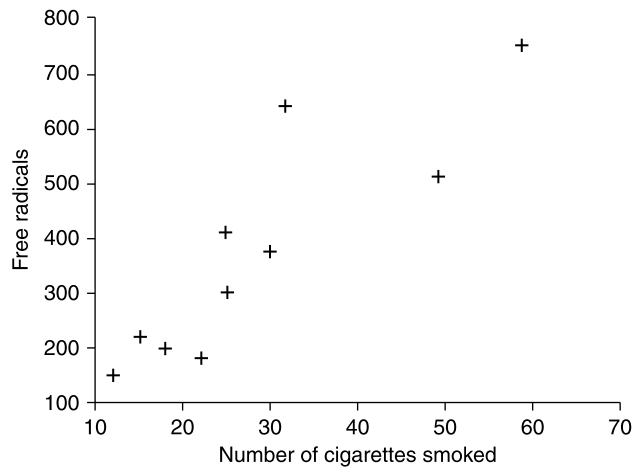


FIGURE 3.9 Cigarettes smoked versus number of free radicals.

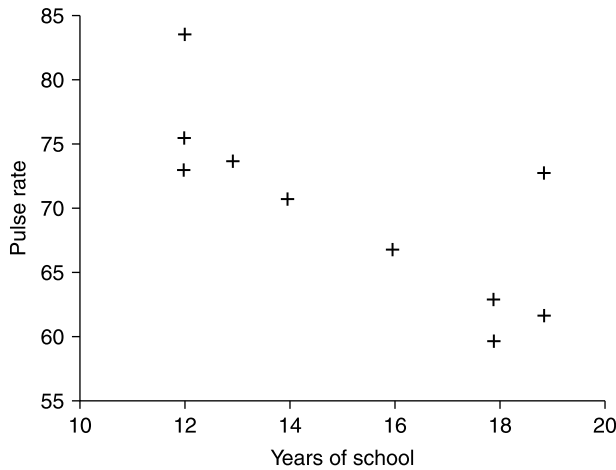
a strong tendency to combine with other atoms within the body.) Figure 3.9 shows the scatter diagram for these data.

From an examination of Fig. 3.9 we see that when the number of cigarettes is high, there tends to be a large number of free radicals, and when the number of cigarettes smoked is low, there tends to be a small number of free radicals. In this case, we say that there is a *positive correlation* between these two variables.

We are also interested in determining the strength of the relationship between a pair of variables in which large values of one variable tend to be associated with small values of the other. For instance, the data of Table 3.4 represent the

Table 3.4 Pulse Rate and Years of School Completed

	Person									
	1	2	3	4	5	6	7	8	9	10
Years of school	12	16	13	18	19	12	18	19	12	14
Pulse rate	73	67	74	63	73	84	60	62	76	71

**FIGURE 3.10** Scatter diagram of years in school and pulse rate.

years of schooling (variable x) and the resting pulse rate in beats per minute (variable y) of 10 individuals. A scatter diagram of this data is presented in Fig. 3.10. From Fig. 3.10 we see that higher numbers of years of schooling tend to be associated with lower resting pulse rates and that lower numbers of years of schooling tend to be associated with the higher resting pulse rates. This is an example of a *negative correlation*.

To obtain a statistic that can be used to measure the association between the individual values of a paired set, suppose the data set consists of the paired values (x_i, y_i) , $i = 1, \dots, n$. Let $\bar{x}$ and $\bar{y}$ denote the sample mean of the x values and the sample mean of the y values, respectively. For data pair i , consider $x_i - \bar{x}$ the deviation of its x value from the sample mean and $y_i - \bar{y}$ the deviation of its y value from the sample mean. Now if x_i is a large x value, then it will be larger than the average value of all the x 's and so the deviation $x_i - \bar{x}$ will be a positive value. Similarly, when x_i is a small x value, then the deviation $x_i - \bar{x}$ will be a negative value. Since the same statements are true about the y deviations, we can conclude the following.

When large values of the x variable tend to be associated with large values of the y variable and small values of the x variable tend to be associated with small values of the y variable, then the signs, either positive or negative, of $x_i - \bar{x}$ and $y_i - \bar{y}$ will tend to be the same.

Now, if $x_i - \bar{x}$ and $y_i - \bar{y}$ both have the same sign (either positive or negative), then their product $(x_i - \bar{x})(y_i - \bar{y})$ will be positive. Thus, it follows that when large x values tend to be associated with large y values and small x values are associated with small y values, then $\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$ will tend to be a large positive number.

The same logic also implies that when large values of one of the variables tend to go along with small values of the other, then the signs of $x_i - \bar{x}$ and $y_i - \bar{y}$ will be opposite, and so $\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$ will be a large negative number.

To determine what it means for $\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$ to be “large,” we standardize this sum first by dividing by $n - 1$ and then by dividing by the product of the two sample standard deviations. The resulting statistic is called the *sample correlation coefficient*.

Definition. Let s_x and s_y denote, respectively, the sample standard deviations of the x values and the y values. The sample correlation coefficient, call it r , of the data pairs (x_i, y_i) , $i = 1, \dots, n$, is defined by

$$\begin{aligned} r &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{(n - 1)s_x s_y} \\ &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} \end{aligned}$$

When $r > 0$, we say that the sample data pairs are positively correlated; and when $r < 0$, we say that they are negatively correlated.

We now list some of the properties of the sample correlation coefficient.

1. The sample correlation coefficient r is always between -1 and $+1$.
2. The sample correlation coefficient r will equal $+1$ if, for some constant a ,

$$y_i = a + bx_i \quad i = 1, \dots, n$$

where b is a positive constant.

3. The sample correlation coefficient r will equal -1 if, for some constant a ,

$$y_i = a + bx_i \quad i = 1, \dots, n$$

where b is a negative constant.

4. If r is the sample correlation coefficient for the data $x_i, y_i, i = 1, \dots, n$, then for any constants a, b, c, d , it is also the sample correlation coefficient for the data

$$a + bx_i, c + dy_i \quad i = 1, \dots, n$$

provided that b and d have the same sign (that is, provided that $bd \geq 0$).

Property 1 says that the sample correlation coefficient r is always between -1 and $+1$. Property 2 says that r will equal $+1$ when there is a straight-line (also called a *linear*) relation between the paired data such that large y values are attached to large x values. Property 3 says that r will equal -1 when the relation is linear and large y values are attached to small x values. Property 4 states that the value of r is unchanged when a constant is added to each of the x variables (or to each of the y variables) or when each x variable (or each y variable) is multiplied by a positive constant. This property implies that r does not depend on the dimensions chosen to measure the data. For instance, the sample correlation coefficient between a person's height and weight does not depend on whether the height is measured in feet or in inches or whether the weight is measured in pounds or kilograms. Also if one of the values in the pair is temperature, then the sample correlation coefficient is the same whether it is measured in degrees Fahrenheit or Celsius.

For computational purposes, the following is a convenient formula for the sample correlation coefficient.

Computational Formula for r

$$r = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sqrt{\left(\sum_{i=1}^n x_i^2 - n\bar{x}^2\right)\left(\sum_{i=1}^n y_i^2 - n\bar{y}^2\right)}}$$

■ Example 3.22

The following table gives the U.S. per capita consumption of whole milk (x) and of low-fat milk (y) in three different years.

	Per capita consumption (gallons)		
	1980	1984	1984
Whole milk (x)	17.1	14.7	12.8
Low-fat milk (y)	10.6	11.5	13.2

Source: U.S. Department of Agriculture, *Food Consumption, Prices, and Expenditures*.

Find the sample correlation coefficient r for the given data.

Solution

To make the computation easier, let us first subtract 12.8 from each of the x values and 10.6 from each of the y values. This gives the new set of data pairs:

	<i>i</i>		
	1	2	3
x_i	4.3	1.9	0
y_i	0	0.9	2.6

Now,

$$\bar{x} = \frac{4.3 + 1.9 + 0}{3} = 2.0667$$

$$\bar{y} = \frac{0 + 0.9 + 2.6}{3} = 1.1667$$

$$\sum_{i=1}^3 x_i y_i = (1.9)(0.9) = 1.71$$

$$\sum_{i=1}^3 x_i^2 = (4.3)^2 + (1.9)^2 = 22.10$$

$$\sum_{i=1}^3 y_i^2 = (0.9)^2 + (2.6)^2 = 7.57$$

Thus,

$$r = \frac{1.71 - 3(2.0667)(1.1667)}{\sqrt{[22.10 - 3(2.0667)^2][7.57 - 3(1.1667)^2]}} = -0.97$$

Therefore, our three data pairs exhibit a very strong negative correlation between consumption of whole and of low-fat milk.

For small data sets such as in [Example 3.22](#), the sample correlation coefficient can be easily obtained by hand. However, for large data sets this computation can become tedious, and a calculator or statistical software is useful. ■

■ Example 3.23

Compute the sample correlation coefficient of the data of [Table 3.3](#), which relates the number of cigarettes smoked to the number of free radicals found in a person's lungs.

Solution

The number of pairs is 10. The pairs are as follows:

18, 202
 32, 644
 25, 411
 60, 755
 12, 144
 25, 302
 50, 512
 15, 223
 22, 183
 30, 375

A calculation shows that the sample correlation coefficient is 0.8759639. ■

The large value of the sample correlation coefficient indicates a strong positive correlation between the number of cigarettes a person smokes and the number of free radicals in that person's lungs.

■ **Example 3.24**

Compute the sample correlation coefficient of the data of [Table 3.4](#), which relates a person's resting pulse rate to the number of years of school completed.

Solution

The pairs are as follows:

12, 73
 16, 67
 13, 74
 18, 63
 19, 73
 12, 84
 18, 60
 19, 62
 12, 76
 14, 71

The sample correlation coefficient is -0.763803 .

The large negative value of the sample correlation coefficient indicates that, for the data set considered, a high pulse rate tends to be associated with

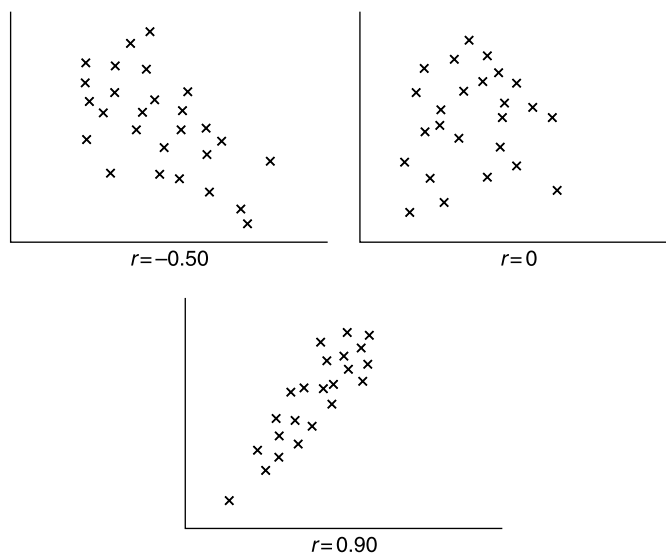
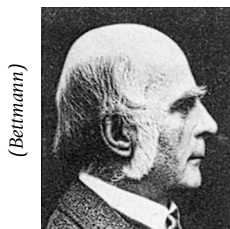


FIGURE 3.11 Sample correlation coefficients.

a small number of years spent in school and a low pulse rate tends to be associated with a large number of years spent in school. ■

The absolute value of the sample correlation coefficient r (that is, $|r|$ —its value without regard to its sign) is a measure of the strength of the linear relationship between the x and the y values of a data pair. A value of $|r|$ equal to 1 means that there is a perfect linear relation; that is, a straight line can pass through all the data points (x_i, y_i) , $i = 1, \dots, n$. A value of $|r|$ of about 0.8 means that the linear relation is relatively strong; although there is no straight line that passes through all the data points, there is one that is “close” to them all. A value of $|r|$ around 0.3 means that the linear relation is relatively weak. The sign of r gives the direction of the relation. It is positive when the linear relation is such that smaller y values tend to go with smaller x values and larger y values with larger x values (and so a straight-line approximation points upward); and it is negative when larger y values tend to go with smaller x values and smaller y values with larger x values (and so a straight-line approximation points downward).

Figure 3.11 displays scatter diagrams for data sets with various values of r .



Francis Galton

Historical Perspective

The development of the concept and utility of the sample correlation coefficient involved the efforts of four of the great men of statistics. The original concept was due to Francis Galton, who was trying to study the laws of inheritance from a quantitative point of view. As such, he wanted to be able to quantify the degree to which characteristics of an offspring relate to those of its

parents. This led him to name and define a form of the sample correlation coefficient that differs somewhat from the one presently in use. Although originally it was meant to be used to assess the hereditary influence of a parent on an offspring, Galton later realized that the sample correlation coefficient presented a method of assessing the interrelation between any two variables.

Although Francis Galton was the founder of the field of biometrics—the quantitative study of biology—its acknowledged leader, at least after 1900, was Karl Pearson. After the Royal Society of London passed a resolution in 1900 stating that it would no longer accept papers that applied mathematics to the study of biology, Pearson, with financial assistance from Galton, founded the statistical journal *Biometrika*, which still flourishes today. The form of the sample correlation coefficient that is presently in use (and that we have presented) is due to Karl Pearson and was originally called *Pearson's product-moment correlation coefficient*.

The probabilities associated with the possible values of the sample correlation coefficient r were discovered, in the case where the data pairs come from a normal population, by William Gosset. There were, however, some technical errors in his derivations, and these were subsequently corrected in a paper by Ronald Fisher.

PROBLEMS

1. Explain why the sample correlation coefficient for the data pairs
 $(121, 360)$, $(242, 362)$, $(363, 364)$
 is the same as that for the pairs
 $(1, 0)$, $(2, 2)$, $(3, 4)$
 which is the same as that for the pairs
 $(1, 0)$, $(2, 1)$, $(3, 2)$
2. Compute the sample correlation coefficient for the data pairs in Prob. 1.

STATISTICS IN PERSPECTIVE

Correlation Measures Association, Not Causation

The results of [Example 3.24](#) indicated a strong negative correlation between an individual's years of education and that individual's resting pulse rate. However, this does not imply that additional years of school will directly reduce one's pulse rate. That is, whereas additional years of school tend to be associated with a lower resting pulse rate, this does not mean that it is a direct *cause* of it. Often the explanation for such an association lies with an unexpressed factor that is related to both variables under consideration. In this instance, it may be that a person who has spent additional time in school is more aware of the latest findings in the area of health and thus may be more aware of the importance of exercise and good nutrition; or perhaps it is not knowledge that is making the difference but rather that people who have had more education tend to end up in jobs that allow them more time for exercise and good nutrition. Probably the strong negative correlation between years in school and resting pulse rate results from a combination of these as well as other underlying factors.

3. The following data represent the IQ scores of 10 mothers and their eldest daughters.

Mother's IQ	Daughter's IQ
135	121
127	131
124	112
120	115
115	99
112	118
104	106
96	89
94	92
85	90

- Draw a scatter diagram.
 - Guess at the value of the sample correlation coefficient r .
 - Compute r .
 - What conclusions can you draw about the relationship between the mother's and daughter's IQs?
4. The following is a sampling of 10 recently released first-time federal prisoners. The data give their crime, their sentence, and the actual time that they served.

Number	Crime	Sentence (months)	Time served (months)
1	Drug abuse	44	24
2	Forgery	30	12
3	Drug abuse	52	26
4	Kidnapping	240	96
5	Income tax fraud	18	12
6	Drug abuse	60	28
7	Robbery	120	52
8	Embezzlement	24	14
9	Robbery	60	35
10	Robbery	96	49

Draw a scatter diagram of the sentence time versus time actually served. Compute the sample correlation coefficient. What does this say about the relationship between the length of a sentence and the time actually served?

- Using the data of Prob. 4, determine the sample correlation coefficient of the sentence time and the proportion of that time actually served. What does this say about the relationship between the length of a sentence and the proportion of this time that is actually served?
- The following data refer to the number of adults in prison and on parole in 12 midwestern states. The data are in thousands of adults.

State	In prison	On parole
Illinois	18.63	11.42
Indiana	9.90	2.80
Iowa	2.83	1.97
Kansas	4.73	2.28
Michigan	17.80	6.64
Minnesota	2.34	1.36
Missouri	9.92	4.53
Nebraska	1.81	0.36
North Dakota	0.42	0.17
Ohio	20.86	6.51
South Dakota	1.05	0.42
Wisconsin	5.44	3.85

- Draw a scatter diagram.
 - Determine the sample correlation coefficient between the number of adults in state prison and on parole in that state.
 - Fill in the missing word. States having a large prison population tend to have a(n) _____ number of individuals on parole.
7. The following data relate the number of criminal cases filed in various U.S. cities to the percentage of those cases that result in a plea of guilty.

City	Percentage of cases resulting in a guilty plea	Number of cases filed
San Diego, CA	73	11,534
Dallas, TX	72	14,784
Portland, OR	62	3892
Chicago, IL	41	35,528
Denver, CO	68	3772
Philadelphia, PA	26	13,796
Lansing, MI	68	1358
St. Louis, MO	63	3649
Davenport, IA	60	1312
Tallahassee, FL	50	2879
Salt Lake City, UT	61	2745

Determine the sample correlation coefficient between the number of cases filed and the percentage of guilty pleas. What can you say about the degree of association between these two variables for these data?

8. The following table gives yearly per capita soft drink consumption (in liters) and the yearly per capita milk consumption (in kg) for a variety of countries. Use it to find the sample correlation coefficient between soft drink and milk consumption.

Per capita soft drink and milk consumption

Country	Soft drink	Milk
United States	216	254
Australia	100	233
Switzerland	81	308
France	37	256
United Kingdom	97	230
The Netherlands	96	329
New Zealand	84	210
Germany	72	314
Italy	50	239
Japan	22	68

9. Find the sample correlation coefficient of the number of hours studied and GPA in Problem 1 of Section 3.5.
10. The following data give the numbers of physicians and dentists, per 100,000 population, in the United States for six different years.

	1980	1981	1982	1983	1985	1986	2001
Physicians	211	217	222	228	237	246	253
Dentists	54	54	55	56	57	57	59

Source: Health Resources Statistics, annual.

- (a) Show that the number of physicians and the number of dentists are positively correlated for these years.
- (b) Do you think that a large value of one of these variables by itself causes a large value of the other? If not, how would you explain the reason for the positive correlation?

The following table gives the death rates by selected causes in different countries. It will be used in Probs. 11 to 13.

Death Rates per 100,000 Population by Selected Causes and Countries

Country	Malignant neoplasm of—						Chronic liver disease and cirrhosis
	Ischemic heart disease	Cerebro-vascular disease	Lung, trachea, bronchus	Stomach	Female breast	Bronchitis, emphysema, asthma	
United States	218.1	60.1	52.7	6.0	31.9	8.3	12.9
Australia	230.9	95.6	41.0	10.1	30.0	16.9	8.7
Austria	155.1	133.2	34.3	20.7	31.6	22.3	26.6
Belgium	120.6	95.0	55.9	14.7	36.8	22.6	12.4
Bulgaria	245.9	254.5	30.6	24.2	21.5	28.6	16.2
Canada	200.6	57.5	50.6	9.0	34.5	9.7	10.1
Czechoslovakia	289.4	194.3	51.3	22.4	27.3	33.8	19.6
Denmark	243.8	73.4	52.2	10.9	39.7	37.1	12.2
Finland	259.8	105.0	36.4	17.3	23.9	19.8	8.8

(Continued)

Country	Malignant neoplasm of—						Chronic liver disease and cirrhosis
	Ischemic heart disease	Cerebro-vascular disease	Lung, trachea, bronchus	Stomach	Female breast	Bronchitis, emphysema, asthma	
France	76.0	79.7	32.2	10.8	27.1	11.7	22.9
Hungary	240.1	186.5	55.0	25.9	31.2	43.8	42.1
Italy	128.9	121.9	42.1	23.9	28.9	30.9	31.5
Japan	41.9	112.8	24.9	40.7	8.1	12.2	14.4
Netherlands	164.6	71.1	56.3	15.6	38.2	17.8	5.5
New Zealand	250.5	98.4	42.0	11.2	37.7	25.8	4.8
Norway	208.5	88.6	26.3	14.4	25.9	18.2	6.9
Poland	109.4	75.3	47.2	24.2	21.1	33.4	12.0
Portugal	76.6	216.4	18.7	26.5	22.6	17.8	30.0
Spain	79.0	133.9	26.0	19.7	19.0	19.1	23.3
Sweden	244.7	73.0	23.2	12.5	26.0	14.3	6.4
Switzerland	112.0	65.6	36.6	12.0	36.6	17.5	10.4
United Kingdom:							
England and Wales	247.6	104.5	57.2	15.2	41.9	24.2	4.8
Scotland	288.0	128.4	68.7	14.9	41.2	14.8	7.3
West Germany	159.5	100.4	34.6	18.3	32.6	26.1	19.3

Source: World Health Organization, *World Health Statistics*.

Use the data on the first ten countries listed to answer Problems 11 through 13.

11. Find the sample correlation coefficient between the death rates of ischemic heart disease and of chronic liver disease.
12. Find the sample correlation coefficient between the death rates of stomach cancer and of female breast cancer.
13. Find the sample correlation coefficient between the death rates of lung cancer and of bronchitis, emphysema, and asthma.
14. In a well-publicized experiment, a University of Pittsburgh researcher enlisted the cooperation of public school teachers in Boston in obtaining a baby tooth from each of their pupils. These teeth were then sawed open and analyzed for lead content. The lead content of each tooth was plotted against the pupil's IQ test score. A strong negative correlation resulted between the amount of lead in the teeth and the IQ scores. Newspapers headlined this result as "proof" that lead ingestion results in decreased scholastic aptitude.
 - (a) Does this conclusion necessarily follow?
 - (b) Offer some other possible explanations.
15. A recent study has found a strong positive correlation between the cholesterol levels of young adults and the amounts of time they spend watching television.
 - (a) Would you have expected such a result? Why?

- (b) Do you think that watching television causes higher cholesterol levels?
 - (c) Do you think that having a high cholesterol level makes a young adult more likely to watch television?
 - (d) How would you explain the results of the study?
16. An analysis relating the number of points scored and fouls committed by basketball players in the Pacific Ten conference has established a strong positive correlation between these two variables. The analyst has gone on record as claiming that this verifies the hypothesis that offensive-minded basketball players tend to be very aggressive and so tend to commit a large number of fouls. Can you think of a simpler explanation for the positive correlation? (*Hint*: Think in terms of the average number of minutes per game that a player is on the court.)
 17. A *New England Journal of Medicine* study published in October 1993 found that people who have guns in their homes for protection are 3 times more likely to be murdered than those with no guns in the home. Does this prove that an individual's chance of being murdered is increased when he or she purchases a gun to keep at home? Explain your answer.
 18. If for each of the fifty states we plot the paired data consisting of the average income of residents of the state and the number of foreign-born immigrants who reside in the state, then the data pairs will have a positive correlation. Can we conclude that immigrants tend to have higher incomes than native-born Americans? If not, how else could this phenomenon be explained?
 19. A study (reported in the May 5, 2008 LA Times) yielded a positive correlation between breast-fed babies and scores on a vocabulary test taken at age 6. Discuss the potential difficulties in interpreting the results of this study.
 20. A study (reported in the March 10, 2009 NY Times) yielded a negative correlation between the age of the father and the results of cognitive tests given to the infant at ages 8 months, 4 years old, and 7 years old. Although the differences in scores between those infants having younger and older fathers was slight the authors of the study called the findings "unexpectedly startling." (On the other hand there was a positive correlation between a mother's age and the cognitive test scores.) Discuss the potential difficulties in interpreting the results of this study.

3.8 THE LORENZ CURVE AND GINI INDEX

The Lorenz curve $L(p)$, $0 \leq p \leq 1$ is a plot related to the incomes of working members of a population, with $L(p)$ representing the proportion of total income that is earned by those in the lower 100p percent of wage earners. For instance, $L(.3)$ is the proportion of total income that is earned by all

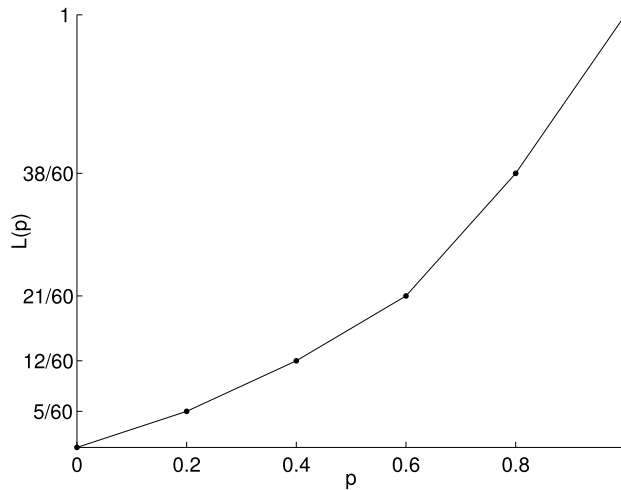


FIGURE 3.12 The Lorenz curve for [Example 3.25](#).

those among the lowest paid 30 percent of the population. In general, suppose there are n working individuals whose incomes are, in increasing order, $x_1 \leq x_2 \leq x_3 \leq \dots \leq x_n$. Because $x_1 + \dots + x_j$ is the total of the incomes of the j lowest paid members of the population, and $x_1 + \dots + x_n$ is the total of all the incomes of the members of the population, it follows that the proportion of all income that is earned by the lowest paid $100j/n$ percent of the population is

$$L(j/n) = \frac{x_1 + \dots + x_j}{x_1 + \dots + x_n}, \quad j = 1, \dots, n$$

It is traditional to let $L(0) = 0$, and to complete the curve by connecting with straight lines between the values $\frac{j}{n}$ and $\frac{j+1}{n}$.

■ Example 3.25

Suppose we want the Lorenz curve when $n = 5$ and the incomes of this group are 9, 7, 22, 5, 17. The incomes in increasing order are 5, 7, 9, 17, 22. Because the sum of all incomes is 60, the Lorenz curve has

$$L(.2) = 5/60$$

$$L(.4) = 12/60$$

$$L(.6) = 21/60$$

$$L(.8) = 38/60$$

$$L(1) = 60/60$$

Connecting with straight lines between these points gives the Lorenz curve ([Fig. 3.12](#)):

Lorenz curves are also used to plot the wealths of members of a population, with $L(p)$ now representing the proportion of all wealth owned by the

least wealthy $100p$ percent of the population. For instance, if the population consists of 1000 people, then $L(.22)$ is the proportion of the population's wealth that is owned by its 220 poorest members. ■

Now, the average of the numbers in a set always increases when a new value that is larger than any of the previous ones is added to the set. For instance, if the initial value is 3 and the successive values to be added are 5, 7, 8 then

$$\begin{aligned} 3 &= 3 \\ \frac{3+5}{2} &= 4 \\ \frac{3+5+7}{3} &= 5 \\ \frac{3+5+7+8}{4} &= 5.75 \end{aligned}$$

Consequently, if the incomes, in increasing value, of the members of the population are $x_1 \leq x_2 \leq x_3 \leq \dots \leq x_n$, then, for any $j = 1, \dots, n$

$$\frac{x_1 + \dots + x_j}{j} \leq \frac{x_1 + \dots + x_{j+1}}{j+1} \leq \dots \leq \frac{x_1 + \dots + x_n}{n}$$

implying that

$$\frac{x_1 + \dots + x_j}{x_1 + \dots + x_n} \leq \frac{j}{n}.$$

Thus,

$$L(j/n) \leq j/n.$$

In addition, it can be verified that $L(j/n) = j/n$, $j = 1, \dots, n$ only when all n incomes are equal. Consequently, unless all incomes are equal, the Lorenz curve is always below the straight line from $(0, 0)$ to $(1, 1)$. (Fig. 3.13 indicates this for the data of Example 3.25).

Thus, the Lorenz curve is equal to the straight line from $(0, 0)$ to $(1, 1)$ when all incomes are equal and dips below the line when incomes are not all equal. The more unequal the incomes the greater the "hump", equal to the region between the straight line and the Lorenz curve (the shaded region in Fig. 3.14).

A measure of the hump and thus of the inequality of the incomes is given by the Gini index, which is the ratio of the area of the hump divided by the area under the straight line. Because the area of a triangle is one half its base times its height, it follows that the area under the straight line from $(0, 0)$ to $(1, 1)$ is equal to $1/2$. Hence, the Gini index, call it G , is given by

$$G = \frac{\text{area of the hump of the Lorenz curve}}{1/2}.$$

Letting B be the area beneath the Lorenz curve, and using that the area of the hump is equal to the area under the straight line minus the area under the

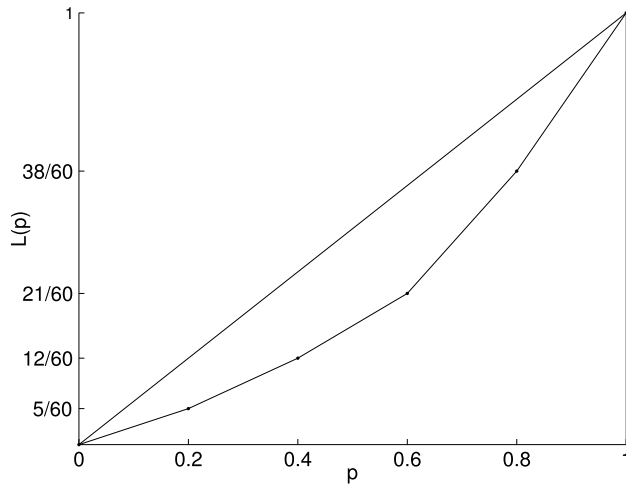


FIGURE 3.13 The Lorenz curve and the straight line.

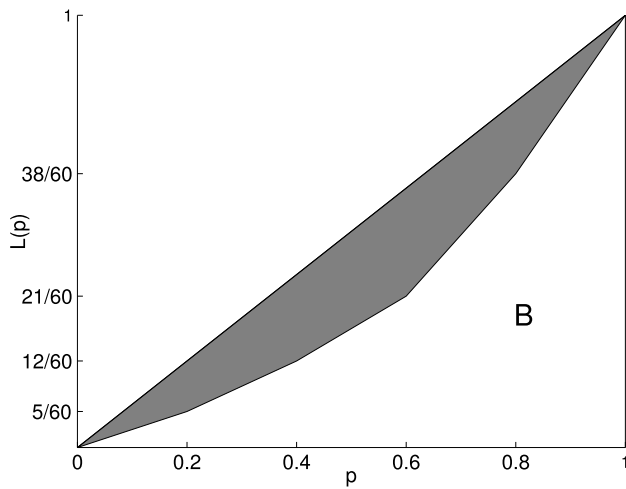


FIGURE 3.14 The hump of the Lorenz curve.

Lorenz curve, we see that the value of the Gini Index is

$$G = \frac{1/2 - B}{1/2} = 1 - 2B.$$

■ Example 3.26

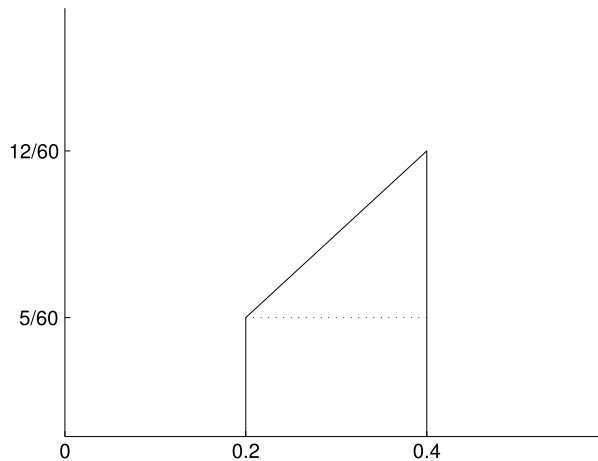
Compute the Gini index for the data of [Example 3.25](#).

Solution

To compute B , the area under the Lorenz curve, we write $B = B_1 + B_2 + B_3 + B_4 + B_5$, where B_1 is the area of the region under the Lorenz curve from 0 to .2; B_2 is the area of the region between .2 and .4; B_3 is the area of the region between .4 and .6; B_4 is the area of the region between .6 and .8; and B_5 is the area of the region between .8 and 1. Now, B_1 is the area of a triangle whose base is .2 and whose height is $5/60$, giving that

$$B_1 = (1/2)(.2)(5/60) = 5/600.$$

Now, B_2 , B_3 , B_4 , and B_5 are the areas of regions composed of a triangle on top of a rectangle. For instance, the following is the region B_2 :



Now all of the triangles in these regions have base .2 and respective heights $5/60$, $7/60$, $9/60$, $17/60$, $22/60$. Consequently, the sum of the areas of all these triangles is

$$\text{Area of Triangles} = \frac{1}{2}(.2)(5 + 7 + 9 + 17 + 22)/60 = .1$$

The 4 rectangles all have base length .2 and respective heights $5/60$, $12/60$, $21/60$, $38/60$, giving that

$$\text{Area of Rectangles} = .2(5 + 12 + 21 + 38) = 60 \approx .25333$$

Adding the area of the rectangle and the area of the triangle gives that $B = .1 + .25333 = .35333$, and thus

$$G = 1 - .70666 = .29334$$

In general, suppose the incomes are, in increasing order, $x_1 \leq x_2 \leq x_3 \leq \dots \leq x_n$. Let

$$s_j = x_1 + \dots + x_j, \quad j = 1, \dots, n$$

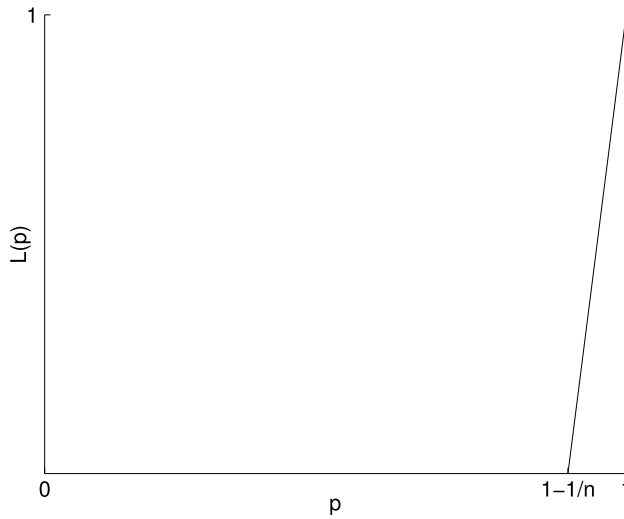


FIGURE 3.15 Largest possible Gini index value.

Then all the triangles will have base $1/n$ and will have respective heights $x_1/s_n, x_2/s_n, \dots, x_n/s_n$. Consequently, the sum of the area of all the triangles is

$$\text{Area of Triangles} = \frac{1}{2n}(x_1 + \dots + x_n)/s_n = \frac{1}{2n}$$

The rectangles will all have length $1/n$ and will have respective heights $s_1/s_n, s_2/s_n, \dots, s_{n-1}/s_n$. Hence,

$$\text{Area of Rectangles} = \frac{s_1 + s_2 + \dots + s_{n-1}}{ns_n},$$

giving that

$$B = \frac{1}{2n} + \frac{s_1 + s_2 + \dots + s_{n-1}}{ns_n}.$$

The Gini index is 0 when all incomes are equal (and so the area between the straight line and the Lorenz curve is 0). On the opposite extreme, the largest possible value of the Gini index occurs when only one of the n members of the population has a positive income. In this case, the area under the Lorenz curve is the area of a triangle whose base is of length $1/n$ and whose height is 1 (see Fig. 3.15). Thus, $B = \frac{1}{2n}$, and $G = 1 - 1/n$.

*3.8.1 The 80–20 and the Pareto Rules

The often quoted “80–20” rule states that 80 percent of the productive work in an organization is done by 20 percent of the workforce, and that 80 percent of

the work done by these productive workers is accomplished by 20 percent of them; and that 80 percent of the work done by these very productive workers is accomplished by 20 percent of them, and so on. When applied to income, it states that the top 20 percent of earners earn 80 percent of the total income of the population; the top 20 percent of the high earners earn 80 percent of the total income of the high earners, and so on. Now, if this is so, then the total income of the bottom 80 percent is only 20 percent of the total income, and so $L(.8) = .2$. Also, because the top 20 percent of the high earners, while comprising $100(.2)(.2) = 4$ percent of all workers, earn $100(.8)(.8) = 64$ percent of the total income, it follows that the bottom 96 percent of the population earns only 36 percent of the total income; that is, $L(.96) = .36$. Continuing on, the rule is equivalent to supposing that the Lorenz curve is such that $L(1 - (.2)^n) = 1 - (.8)^n$, $n \geq 1$.

The 80–20 rule holds when the distribution of income in a population follows a special case of the Pareto rule. Consider a population of size n and suppose that the minimal annual income among this group is α : Letting $P(y)$ denote the proportion of the population that earns at least y , then we say income distribution follows a *Pareto rule* with parameters α and λ if for some $\lambda > 1$

$$P(y) = \left(\frac{\alpha}{y}\right)^\lambda, \quad y > \alpha$$

It can be shown that the average income of the members of this population is

$$\text{average income} = \alpha \frac{\lambda}{\lambda - 1}. \quad (3.3)$$

Thus, for instance, if $\alpha = 20,000$ and $\lambda = 2$, then the proportion of the population that earns at least 40,000 is $1/4$ (that is, 25 percent of the population earns over 40,000) and the proportion that earns at least 60,000 is $1/9$ (meaning that 100/9 percent of the population earns at least that amount). Also, 40,000 is the average income of members of this population.

If we let $N(y)$ denote the number of members of the population whose income is at least y , then

$$N(y) = nP(y) = n\left(\frac{\alpha}{y}\right)^\lambda, \quad y > \alpha.$$

Now, for a given value $x > \alpha$, let $S(x)$ denote the subpopulation consisting of all the people in the population that earn at least x . Letting, for $y > x$, $P_x(y)$ denote the proportion of the members of $S(x)$ that earn at least y ; then

$$P_x(y) = \frac{\text{number in } S(x) \text{ earning at least } y}{\text{number in } S(x)}, \quad y > x$$

Because, for $y > x$, the number of people in the subpopulation that earn at least y is equal to the number of people in the entire population that earn at

least y , the preceding gives that

$$P_x(y) = \frac{N(y)}{N(x)} = \left(\frac{x}{y}\right)^\lambda, \quad y > x$$

In other words, the distribution of income for the subpopulation that earn at least x follows a Pareto rule with parameter α being changed to x . Thus, from (3.3), it follows that

$$\text{average income of those in } S(x) = x \frac{\lambda}{\lambda - 1} \quad (3.4)$$

Now, let $F(x)$ denote the fraction of all income that is earned by those in $S(x)$. Because there are $nP(x)$ individuals in $S(x)$, with an average income of $x \frac{\lambda}{\lambda - 1}$ it follows that the total income earned by those in $S(x)$ is $nP(x)x \frac{\lambda}{\lambda - 1}$. As the total income earned by all members of the population is $n\alpha \frac{\lambda}{\lambda - 1}$, we obtain that

$$F(x) = \frac{nP(x)x \frac{\lambda}{\lambda - 1}}{n\alpha \frac{\lambda}{\lambda - 1}} = \frac{P(x)x}{\alpha} = \frac{x}{\alpha} \left(\frac{\alpha}{x}\right)^\lambda \quad (3.5)$$

Now, if x_0 is such that 20 percent of the population earns at least x_0 , then

$$\frac{1}{5} = P(x_0) = \left(\frac{\alpha}{x_0}\right)^\lambda,$$

which will be true if

$$x_0 = \alpha 5^{1/\lambda} \quad (3.6)$$

The fraction of all income that is earned by members of $S(x_0)$ is, from (3.5),

$$F(x_0) = \frac{x_0}{\alpha} \left(\frac{\alpha}{x_0}\right)^\lambda = 5^{1/\lambda} \frac{1}{5}$$

Therefore, the top 20 percent will earn 80 percent of the total income if

$$.8 = F(x_0)$$

which is equivalent to

$$\frac{4}{5} = 5^{1/\lambda} \frac{1}{5}$$

Taking logarithms of both sides of the preceding gives that

$$\lambda = \log(5) / \log(4)$$

Thus, if income follows a Pareto rule with $\lambda = \log(5) / \log(4) \approx 1.161$, then the top 20 percent of high earners will earn 80 percent of the total income. Also, it can be shown that the top 20 of these high earners – who constitute the top 4 percent of all earners – will earn 80 percent of all income of the high earners (which constitutes 64 percent of all earnings), and so on.

PROBLEMS

1. Draw the Lorenz curve of a group whose incomes are 25, 32, 60, 40, 38, 50.
2. Draw the Lorenz curve of a group whose annual incomes (in thousands of dollars) are given by the following frequency table.

Value	Frequency
30	2
50	4
60	5
90	4
100	3
120	2

3. Compute the Gini index for wealth of a population of size 5 with wealths 40, 50, 55, 200, 320.
4. Compute the Gini index for the group of Problem 1.
5. Compute the Gini index for the incomes given by the following frequency table:

Value	Frequency
10	2
14	1
20	3

6. Let G be the Gini index of the wealths of members of a certain population. Suppose the members of the population can be divided into two groups. If G_1 is the Gini index for data consisting of members of the first group, and G_2 is the Gini index for data consisting of members of the second group, is $G = pG_1 + (1 - p)G_2$, where p is the proportion of the population that is in the first group.
7. Is the Gini index changed if a constant is added to all values.
8. Is the Gini index changed if all values are multiplied by a positive constant.
9. *If a population's weekly income follows the Pareto rule with $\alpha = 500$, $\lambda = 2$, what fraction of the population earns above 1100.
10. *Suppose that a population's monthly income follows the Pareto rule with $\alpha = 1,500$, $\lambda = 3$.
 - (a) What fraction of the population earns less than 2500.
 - (b) What fraction of the population earns between 2300 and 3000.

3.9 USING R

To find the sample mean and sample variance of data values. Suppose you want the sample mean and sample variance of the values $x_1, \dots, x_n$. Type

$$x = c(x_1, \dots, x_n)$$

and hit return. [Interpret $x = c(x_1, \dots, x_n)$ as meaning that x is the sequence of values $x_1, \dots, x_n$.] On the next line type

```
mean(x)
```

and when you hit return, the sample mean is displayed. On the next line, type

```
var(x)
```

and when you hit return, the variance is displayed.

For instance, suppose you wanted the sample mean and sample variance of the values 4, 6, 12, 9, 21, 14. Then

```
> x = c(4, 6, 12, 9, 21, 14)
> mean(x)
[1] 11
> var(x)
[1] 37.6
```

By the way, you do not type `>` at the beginning of a line. R always starts a line with this symbol, which is a prompt asking what to do next. (The `[1]` in front of the answers returned is to indicate that you only asked for a single item.)

Other commands are `sum(x)`, which returns the sum of the values; `median(x)`, which returns the sample median; and `sd(x)`, which returns the sample standard deviation (equal to the square root of `var(x)`).

If you have data pairs $(x_1, y_1), \dots, (x_n, y_n)$, then to obtain the sample correlation coefficient, do the following

```
> x = c(x_1, ..., x_n)
> y = c(y_1, ..., y_n)
> cor(x, y)
```

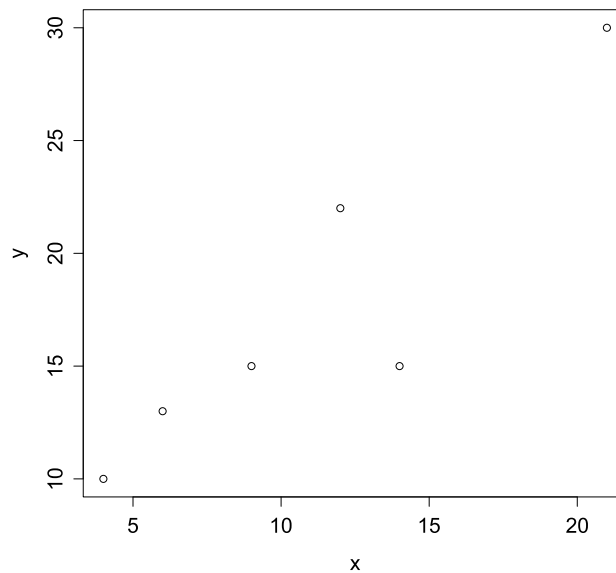
To obtain a scatter plot of the data, type

```
> plot(x, y)
```

and hit return.

For instance, suppose we have the data pairs (4, 10), (6, 13), (12, 22), (9, 15), (21, 30), (14, 15), then we obtain the sample correlation coefficient and scatter diagram as follows:

```
> x = c(4, 6, 12, 9, 21, 14)
> y = c(10, 13, 22, 15, 30, 15)
> cor(x, y)
[1] 0.9041494
> plot(x, y)
```



R can also be used for computation. For instance, suppose we wanted to compute $\frac{18\sqrt{177}}{677}$. This is accomplished thusly:

```
> 18 * sqrt(177)/677
[1] 0.3537288
```

KEY TERMS

Statistic: A numerical quantity whose value is determined by the data.

Sample mean: The arithmetic average of the values in a data set.

Deviation: The difference between the individual data values and the sample mean. If x_i is the i th data value and $\bar{x}$ is the sample mean, then $x_i - \bar{x}$ is called the i th deviation.

Sample median: The middle value of an ordered set of data. For a data set of n values, the sample median is the $(n + 1)/2$ -smallest value when n is odd and the average of the $n/2$ - and $n/2 + 1$ -smallest values when n is even.

Sample $100p$ percentile: That data value such that at least $100p$ percent of the data are less than or equal to it and at least $100(1 - p)$ percent of the data are greater than or equal to it. If two data values satisfy this criterion, then it is the average of them.

First quartile: The sample 25th percentile.

Second quartile: The sample 50th percentile, which is also the sample median.

Third quartile: The sample 75th percentile.

Sample mode: The data value that occurs most frequently in a data set.

Sample variance: The statistic s^2 , defined by

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

It measures the average of the squared deviations.

Sample standard deviation: The positive square root of the sample variance.

Range: The largest minus the smallest data value.

Interquartile range: The third quartile minus the first quartile.

Normal data set: One whose histogram is symmetric about its middle interval and decreases on both sides of the middle in a bell-shaped manner.

Skewed data set: One whose histogram is not symmetric about its middle interval. It is said to be skewed to the right if it has a long tail to the right and skewed to the left if it has a long tail to the left.

Bimodal data set: One whose histogram has two local peaks or humps.

Sample correlation coefficient: For the set of paired values $x_i, y_i, i = 1, \dots, n$, it is defined by

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{(n - 1)s_x s_y}$$

where $\bar{x}$ and s_x are, respectively, the sample mean and the sample standard deviation of the x values, and similarly for $\bar{y}$ and s_y . A value of r near $+1$ indicates that larger x values tend to be paired with larger y values and smaller x values tend to be paired with smaller y values. A value near -1 indicates that larger x values tend to be paired with smaller y values and smaller x values tend to be paired with larger y values.

Lorenz curve: A plot of the fraction of the total income earned by members of a population that is earned by the lowest paid $100p$ percent of the population.

SUMMARY

We have seen three different statistics which describe the center of a data set: the sample mean, sample median, and sample mode.

The sample mean of the data $x_1, \dots, x_n$ is defined by

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

and is a measure of the center of the data.

If the data are specified by the frequency table

Value	Frequency
x_1	f_1
x_2	f_2
$\vdots$	$\vdots$
x_k	f_k

then the sample mean of the $n = \sum_{i=1}^k f_i$ data values can be expressed as

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{n}$$

A useful identity is

$$\sum_{i=1}^n (x_i - \bar{x}) = 0$$

The sample median is the middle value when the data are arranged from smallest to largest. If there are an even number of data points, then it is the average of the two middle values. It is also a measure of the center of the data set.

The sample mode is that value in the data set that occurs most frequently.

Suppose a data set of size n is arranged from smallest to largest. If np is not an integer, then the sample $100p$ percentile is the value whose position is the smallest integer larger than np . If np is an integer, then the sample $100p$ percentile is the average of the values in positions np and $np + 1$.

The sample 25th percentile is the *first quartile*. The sample 50th percentile (which is equal to the sample median) is called the *second quartile*, and the sample 75th percentile is called the *third quartile*.

The sample variance s^2 is a measure of the spread in the data and is defined by

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

where n is the size of the set. Its square root s is called the *sample standard deviation*, and it is measured in the same units as the data.

The following identity is useful for computing the sample variance by using pencil and paper or a hand calculator.

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2$$



Program 3-1 will compute the sample mean, sample variance, and sample standard deviation of any set of data.

Another statistic that describes the spread of the data is the *range*, the difference between the largest and smallest data values.

Normal data sets will have their sample mean and sample median approximately equal. Their histograms are symmetric about the middle interval and exhibit a bell shape.

The sample correlation coefficient r measures the degree of association between two variables. Its value is between -1 and $+1$. A value of r near $+1$ indicates that when one of the variables is large, the other one also tends to be large and that when one of them is small, the other also tends to be small. A value of r near -1 indicates that when one of the variables is large, the other one tends to be small.

A large value of $|r|$ indicates a strong association between the two variables. Association does not imply causation.

The Lorenz curve is a plot of $L(p)$, $0 \leq p \leq 1$, where $L(p)$ is equal to the proportion of the total income of a population that is earned by the lowest paid $100p$ percent. It is never above the straight line from $(0, 0)$ to $(1, 1)$. The Gini index, defined as twice the area between the straight line and the Lorenz curve, is often used as an indication of how much inequality there is in the incomes of members of the population. For instance, when everyone earns the same amount, the Lorenz curve matches the straight line, and so the Gini index is 0.

REVIEW PROBLEMS

- Construct a data set that is symmetric about 0 and contains
 - Four distinct values
 - Five distinct values
 - In both cases, compute the sample mean and sample median.
- The following stem-and-leaf plot records the diastolic blood pressure of a sample of 30 men.

9	3, 5, 8
8	6, 7, 8, 9, 9, 9
7	0, 1, 2, 2, 4, 5, 5, 6, 7, 8
6	0, 1, 2, 2, 3, 4, 5, 5
5	4, 6, 8

- Compute the sample mean $\bar{x}$.
 - Compute the sample median.
 - Compute the sample mode.
 - Compute the sample standard deviation s .
 - Do the data appear to be approximately normal?
 - What proportion of the data values lies between $\bar{x} + 2s$ and $\bar{x} - 2s$?
 - Compare the answer in part (f) to the one prescribed by the empirical rule.
- The following data are the median ages of residents in each of the 50 states of the United States:

29.3	27.7	30.4	31.1	28.5
32.1	28.0	31.3	26.6	25.8
25.9	33.0	31.5	30.0	28.4
24.9	31.6	26.6	25.4	29.2
29.3	27.9	31.8	31.5	30.3
28.5	29.3	26.6	31.2	32.1
31.4	30.1	27.0	28.5	27.6
28.9	29.4	30.5	31.2	29.4
29.3	30.1	28.8	27.9	30.4
32.3	30.4	25.8	27.1	26.9

- (a) Find the median of these ages.
 - (b) Is this necessarily the median age of all people in the United States? Explain.
 - (c) Find the quartiles.
 - (d) Find the sample 90th percentile.
4. Use Table 3.2 in Example 3.19 to fill in the answers.
- (a) To have one's score be among the top 10 percent of all physical science students, it must be at least ____.
 - (b) To have one's score be among the top 25 percent of all social science students, it must be at least ____.
 - (c) To have one's score be among the bottom 50 percent of all medical students, it must be less than or equal to ____.
 - (d) To have one's score be among the middle 50 percent of all law school students, it must be between ____ and ____.
5. The number of violent offenses per 100,000 population is given here for each of the 50 states. Is this data set approximately normal?

Violent Crime per 100,000 Population, 2002

State	Rate	Rank	State	Rate	Rank
United States	495	(X)	Montana	352	27
Alabama	444	21	Nebraska	314	32
Alaska	563	12	Nevada	638	7
Arizona	553	13	New Hampshire	161	47
Arkansas	424	22	New Jersey	375	25
California	593	10	New Mexico	740	4
Colorado	352	27	New York	496	17
Connecticut	311	33	North Carolina	470	19
Delaware	599	9	North Dakota	78	50
Florida	770	2	Ohio	351	29
Georgia	459	20	Oklahoma	503	16
Hawaii	262	41	Oregon	292	34
Indiana	357	26	Pennsylvania	402	23
Idaho	255	42	Rhode Island	285	37
Illinois	621	8	South Carolina	822	1
Iowa	286	36	South Dakota	177	46
Kansas	377	24	Tennessee	717	5
Kentucky	279	38	Texas	579	11
Louisiana	662	6	Utah	237	43
Maine	108	48	Vermont	107	49
Maryland	770	2	Virginia	291	35
Massachusetts	484	18	Washington	345	30
Michigan	540	14	West Virginia	234	44
Minnesota	268	40	Wisconsin	225	45
Mississippi	343	31	Wyoming	274	39
Missouri	539	15			

Note Violent crime refers to violent offenses known to the police, which includes murder, forcible rape, robbery, and aggravated assault. When states share the same rank, the next lower rank is omitted. Because of rounded data, states may have identical values shown but different ranks.

6. The following data represent the birth weights at an inner-city hospital in a large eastern city:

2.4, 3.3, 4.1, 5.0, 5.1, 5.2, 5.6, 5.8, 5.9, 5.9, 6.0, 6.1, 6.2, 6.3,
6.3, 6.4, 6.4, 6.5, 6.7, 6.8, 7.2, 7.4, 7.5, 7.5, 7.6, 7.6, 7.7, 7.8,
7.8, 7.9, 7.9, 8.3, 8.5, 8.8, 9.2, 9.7, 9.8, 9.9, 10.0, 10.3, 10.5

- Plot this in a stem-and-leaf diagram.
 - Find the sample mean $\bar{x}$.
 - Find the sample median.
 - Find the sample standard deviation s .
 - What proportion of the data lies within $\bar{x} \pm 2s$?
 - Do the data appear to be approximately normal?
 - If your answer to (f) is yes, what would you have estimated, based on your answers to (b) and (d), for (e)?
- *7. Let a and b be constants. Show that if $y_i = a + bx_i$ for $i = 1, \dots, n$, then r , the sample correlation coefficient of the data pairs $x_i, y_i, i = 1, \dots, n$, is given by
- $r = 1$ when $b > 0$
 - $r = -1$ when $b < 0$
- (Hint: Use the definition of r , not its computational formula.)
8. The following data are taken from the book *Researches on the Probability of Criminal and Civil Verdicts*, published in 1837 by the French mathematician and probabilist Simeon Poisson. The book emphasized legal applications of probability. The data refer to the number of people accused and convicted of crimes in France from 1825 to 1830.

Year	Number accused	Number convicted
1825	6652	4037
1826	6988	4348
1827	6929	4236
1828	7396	4551
1829	7373	4475
1830	6962	4130

- Determine the sample mean and sample median of the number accused.
- Determine the sample mean and sample median of the number convicted.
- Determine the sample standard deviation of the number accused.
- Determine the sample standard deviation of the number convicted.
- Would you expect the number accused and the number convicted to have a positive or a negative sample correlation coefficient?
- Determine the sample correlation coefficient of the number of accused and number of convicted.
- Determine the sample correlation coefficient between the number accused and the percentage of these who are convicted.

- (h) Draw scatter diagrams for parts (f) and (g).
 - (i) Guess at the value of the sample correlation coefficient between the number of convicted and the percentage convicted.
 - (j) Draw a scatter diagram for the variables in (i).
 - (k) Determine the sample correlation coefficient for the variables in part (i).
9. Recent studies have been inconclusive about the connection between coffee consumption and coronary heart disease. If a study indicated that consumers of large amounts of coffee appeared to have a greater chance of suffering heart attacks than did drinkers of moderate amounts or drinkers of no coffee at all, would this necessarily “prove” that excessive coffee drinking leads to an increased risk of heart attack? What other explanations are possible?
 10. Recent studies have indicated that death rates for married middle-aged people appear to be lower than for single middle-aged people. Does this mean that marriage tends to increase one’s life span? What other explanations are possible?
 11. A June 9, 1994, article in *The New York Times* noted a study showing that years with low inflation rates tend to be years with high average-productivity increases. The article claimed that this supported the Federal Reserve Board’s claim that a low rate of inflation tends to result in an increase in productivity. Do you think the study provides strong evidence for this claim? Explain your answer.
 12. The sample mean of the initial 99 values of a data set consisting of 198 values is equal to 120, whereas the sample mean of the final 99 values is equal to 100.
 - (a) What can you conclude about the sample mean of the entire data set
 - (b) Repeat when “sample mean” is replaced by “sample median”.
 - (c) Repeat when “sample mean” is replaced by “sample mode”.
 13. Determine the Gini index of the incomes of a population of size 6 whose individual incomes are 33, 45, 70, 62, 40, 55.
 14. A random sample of individuals were rated as to their standing posture. In addition, the numbers of days of back pain each had experienced during the past year were also recorded. Surprisingly to the researcher these data indicated a positive correlation between good posture and number of days of back pain. Does this indicate that good posture causes back pain?
 15. The following are the number of traffic deaths in a sample of states, both for 2007 and 2008. Plot a scatter diagram and find the sample correlation coefficient for the data pairs.

2007 and 2008 Traffic Fatalities per State

State	2007	2008
WY	149	159
IL	1248	1044
MA	434	318
NJ	724	594
MD	615	560
OR	452	414
WA	568	504
FL	3221	2986
UT	291	271
NH	129	139